

Prediction-Powered Data Fusion for Treatment Effect Estimation

Yonghan Jung

Working draft, August 28, 2026

Randomized controlled trials identify causal effects under transparent design assumptions, but their sample sizes can make average and conditional treatment-effect estimation imprecise. Observational studies are often larger and can yield accurate outcome predictions, but those predictions need not be causal. We develop a model-agnostic prediction-powered framework that keeps the randomized trial as the causal anchor while using observational outcome models and covariates through two tunable channels. The coefficient λ changes the outcome regression inside an RCT-valid pseudo-outcome, and ω introduces a prediction-powered control variate centered across the two samples.

For the average treatment effect, we derive exact conditional unbiasedness, a within-/between-covariate variance decomposition, the joint population variance minimizer, its population-orthogonal simplification, singular and constrained cases, and an honest plug-in algorithm. A separate conditional central limit theorem supports Wald inference after independent coefficient selection, while nested outer cross-fitting requires additional stability analysis for variance estimation. For conditional effects, we formulate a prediction-powered squared-loss learner from an RCT-valid AIPW pseudo-outcome, distinguish pseudo-outcome-noise, fixed-loss-variance, and final learner-risk oracles, and give implementable linear-sieve and frozen-neural-basis versions with honest common-reference selection. We outline an R-loss analogue only as a prospective extension requiring a separate derivation. Exact density-ratio transport covers unequal covariate marginals; classifier odds, Riesz learning, and loss-targeted balancing enter an estimated-weight regime with explicit drift. Existing simulations support the ATE construction in several small-trial regimes. CATE experiments and real-outcome analyses remain planned, so neither the theory nor the current evidence establishes universal finite-sample improvement for estimated coefficients.

Contents

1. Introduction	3
1.1. Categorized literature review	3
2. Background and Problem Setting	4
2.1. Prediction-powered inference and PPI++	4
2.2. RCT-OBS problem setting and notation	4

2.3. Existing data-fusion estimators	6
3. Prediction-Powered Fusion for the ATE	7
3.1. Estimator and exact variance	7
3.2. Joint and separable population oracles	8
3.3. Estimable honest algorithm	11
3.3.1. Coefficient estimation and honest evaluation	11
3.3.2. Estimating the density ratio	14
3.3.3. Optional nested cross-fitting	18
3.3.4. Diagnostics	19
4. Prediction-Powered Fusion Learning for CATE	19
4.1. CATE targets and RCT-valid learning losses	19
4.2. Orthogonality audit	22
4.3. Final learner-risk coefficients and honest selection	23
4.4. Why general error analysis is hard and the linear-sieve solution	24
4.5. Implementation boundaries and open theory	26
5. Experiments	26
5.1. Completed ATE benchmarks	26
5.2. Planned CATE evaluation and interpretation	28
5.3. Scope, failure modes, and conclusion	28
A. Simulation Details	31
B. Proof Details	33
B.1. Conditional validity of the trial pseudo-outcome	33
B.2. Proof of Corollary 1	33
B.3. Proof of Theorem 1	33
B.4. Proof of Proposition 2	34
B.5. Proof of Proposition 3	34
B.6. Proof of Theorem 2	34
B.7. Singular and constrained extensions	34
B.8. Proof of Corollary 2.1	35
B.9. Proof of Corollary 2.2	36
B.10. Prediction-powered CATE losses and finite-sample theory	36
B.11. A two-stratum oracle calculation	38
B.12. Honest variance estimation and evaluation-sample inference	38
B.13. Proof of Theorem 4	39
B.14. Density-ratio characterizations and estimated-ratio inheritance	39
B.15. Outer-fold honesty	41
B.16. Counterexamples to adaptive reuse	41

1. Introduction

1.1. Categorized literature review

Prediction-powered inference. Prediction-powered inference uses a large set of machine predictions together with a smaller labeled sample, correcting the prediction-based estimate with labeled residuals so that the statistical target does not require the predictor to be correct (Angelopoulos et al., 2023). This separation between predictive quality and inferential validity is the starting point for the present construction. Randomized observations play the role of causally valid labeled data, while observational outcome models supply potentially useful predictions.

Adaptive prediction-powered inference. PPI++ introduces data-adaptive power tuning that adjusts how strongly predictions enter an estimating equation and reports computational and statistical gains over the original construction (Angelopoulos et al., 2024). Its central lesson is that a prediction should be used in proportion to its empirical value rather than being accepted as truth. Our coefficient ω follows this control-variate logic, but the target and score are causal quantities identified by the randomized design.

Causal data fusion and transport. General causal data-fusion theory distinguishes data collected under different regimes and populations and asks whether a target causal query is identified from their combination (Bareinboim and Pearl, 2016). Reviews of causal data integration likewise separate randomized trials, observational studies, historical controls, and heterogeneous target populations (Shi et al., 2023). The present paper assumes that the trial already identifies the target effect and studies an estimation question: how can observational predictions improve precision without replacing randomized identifying information?

Combining randomized and observational estimates. Existing estimators combine experimental and observational effect estimates through shrinkage, bias detection, or adaptive weighting. Shrinkage estimators can reduce risk when observational bias is sufficiently controlled (Rosenman et al., 2023), while adaptive combination methods move toward the randomized estimator when observational bias is detected (Cheng and Cai, 2021). These approaches combine effect estimates from the two sources. Our construction instead inserts the observational outcome model into an RCT-valid pseudo-outcome and separately centers a prediction-based covariate statistic.

CATE learners and orthogonal losses. Meta-learners convert standard regression algorithms into conditional effect estimators (Künzel et al., 2019). The R-learner residualizes outcomes and treatments to isolate treatment-effect heterogeneity (Nie and Wager, 2021), while doubly robust pseudo-outcome regression can retain favorable error structure under nuisance estimation (Kennedy, 2023). Direct multi-source learners have also been proposed for heterogeneous causal data fusion (Li et al., 2022). These results motivate treating the prediction-powered construction as a loss wrapper that can be paired with a linear sieve or a fixed learned representation.

Covariate shift and balancing. When trial and observational covariate marginals differ, exact density-ratio weighting transports fixed covariate functions between populations. Riesz-representer methods provide an optimization perspective on such linear functionals and on debiasing (Chernozhukov et al., 2021). Density-ratio estimation, generic balancing, and Riesz learning are established tools. The project-specific question is whether they can be targeted to the generated CATE loss class $\{(g - t)^2 : t \in \mathcal{T}\}$ while preserving an RCT-defined risk; we treat this specialization as a working research direction rather than a settled novelty claim.

2. Background and Problem Setting

2.1. Prediction-powered inference and PPI++

Let (X_i, Y_i) , $i = 1, \dots, n$, be labeled pairs and let $\tilde{X}_j$, $j = 1, \dots, N$, be unlabeled covariates. Suppose a fixed predictor supplies $f(X_i)$ and $f(\tilde{X}_j)$. For $\omega \in \mathbb{R}$, the PPI++ estimator (Angelopoulos et al., 2024) of the labeled-population mean is

$$\hat{\mu}_{\text{PPI}++}(\omega) \triangleq \frac{1}{n} \sum_{i=1}^n Y_i + \omega \left\{ \frac{1}{N} \sum_{j=1}^N f(\tilde{X}_j) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right\}. \quad (1)$$

When labeled and unlabeled covariates share a marginal law and the two samples are independent, the expression in braces has mean zero for every fixed predictor f . PPI++ selects ω to control how strongly this mean-zero prediction difference enters the estimator. Our construction retains this control-variate role and introduces a separate outcome-regression blending coefficient only after defining the RCT-valid score.

2.2. RCT-OBS problem setting and notation

Let $\mathcal{K} \triangleq \{\text{eval}, \text{nuis}, \text{tune}\}$. For each source $S \in \{R, O\}$ and role $k \in \mathcal{K}$, observe

$$D_S^k \triangleq \{V_{S,i}^k = (X_{S,i}^k, A_{S,i}^k, Y_{S,i}^k) : i = 1, \dots, n_S^k\} \sim P_S^{\otimes n_S^k}. \quad (2)$$

Here X takes values in the covariate space $\mathcal{X}$, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. For each $S \in \{R, O\}$, the datasets D_S^{eval} , D_S^{nuis} , and D_S^{tune} are used for final evaluation, nuisance and density-ratio fitting, and coefficient or candidate tuning, respectively. Set $n \triangleq n_R^{\text{eval}}$ and $N \triangleq n_O^{\text{eval}}$, and define $\mathbb{P}_{R,n}h \triangleq n^{-1} \sum_i h(V_{R,i}^{\text{eval}})$ and $\mathbb{P}_{O,N}h \triangleq N^{-1} \sum_j h(V_{O,j}^{\text{eval}})$. Every displayed moment is finite; square-integrability and stronger moment conditions are stated locally when variance estimation or normal-approximation inference requires them.

Let $Y(1)$ and $Y(0)$ denote potential outcomes. The RCT-population conditional and average treatment effects are

$$\tau(x) \triangleq \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}, \quad \theta \triangleq \mathbb{E}_R\{\tau(X)\}. \quad (3)$$

Assumption 1 (Trial identification). Consistency holds, $\{Y(0), Y(1)\} \perp A \mid X$ under P_R , and the known trial propensity $e(x) \triangleq P_R(A = 1 \mid X = x)$ satisfies $\epsilon < e(X) < 1 - \epsilon$ almost surely for a constant $\epsilon > 0$.

We study both the common-covariate regime $P_R^X = P_O^X$, formalized in Assumption 2, and the more general covariate-shift regime $P_R^X \neq P_O^X$, formalized in Assumption 3.

Definition 1 (Target-to-source density ratio). Whenever $P_R^X \ll P_O^X$, define the target-to-

source density ratio by

$$r_0(x) \triangleq \frac{dP_R^X}{dP_O^X}(x).$$

It is the Radon–Nikodym derivative that transports an OBS covariate expectation to the RCT covariate population.

Assumption 2 (Common covariate marginal). The trial and observational covariates have the same marginal law, $P_R^X = P_O^X$. This assumption does not equate the treatment or outcome mechanisms across sources, and the observational treatment contrast may remain confounded. In this regime, Definition 1 gives $r_0(x) = 1$ for P_O^X -almost every x .

Assumption 3 (Covariate shift and support overlap). Under covariate shift, $P_R^X \ll P_O^X$, so the RCT covariate population has no support outside the OBS covariate population. Definition 1 therefore defines r_0 . This assumption establishes existence of the ratio; it does not assert that r_0 is known or exactly estimated.

Assumptions 2 and 3 describe alternative covariate regimes, not cumulative restrictions. Under Assumption 2, no transport weighting is needed and we set $r \triangleq 1$. When the covariate marginals differ, Assumption 3 is needed to convert OBS covariate expectations into expectations under the RCT population. Neither regime assumes that the OBS treatment contrast is causal. Exact-oracle results below use $r \triangleq r_0$. Feasible results may replace r_0 by an evaluation-independent estimate $\hat{r}$ under the conditions in Section 3.3.2.

Assumption 4 (Honest sample separation). The nuisance samples $D_R^{\text{nuis}}, D_O^{\text{nuis}}$, the tuning samples $D_R^{\text{tune}}, D_O^{\text{tune}}$, and the evaluation samples $D_R^{\text{eval}}, D_O^{\text{eval}}$ are mutually independent. Nuisance and density-ratio estimators use only $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. Coefficients and candidate rules use only $D_R^{\text{tune}}, D_O^{\text{tune}}$ after the nuisance objects are fixed. If candidate fitting and validation both use tuning data, their assignments are prespecified and disjoint. Final point estimation, variance estimation, and normal-approximation inference use only $D_R^{\text{eval}}, D_O^{\text{eval}}$, after every learned object and split assignment has been fixed.

Throughout, every P_R - or P_O -expectation, covariance, variance, and exact variance result that involves a learned object is conditional on the realized nuisance and tuning datasets, split assignments, and pre-evaluation algorithmic randomness. These objects are held fixed, and only the evaluation observations remain random. We suppress this conditioning in the notation.

For $S \in \{R, O\}$ and $a \in \{0, 1\}$, define the population outcome-regression target $\mu_{S,a} : \mathcal{X} \rightarrow \mathbb{R}$ by

$$\mu_{S,a}(x) \triangleq \mathbb{E}_S[Y \mid X = x, A = a].$$

For each $S \in \{R, O\}$ and $a \in \{0, 1\}$, let $\hat{\mu}_{S,a} : \mathcal{X} \rightarrow \mathbb{R}$ denote a fixed pre-evaluation predictor of the corresponding population regression $\mu_{S,a}$. The RCT predictor is trained only on D_R^{nuis} , and the OBS predictor is trained only on D_O^{nuis} . No correctness assumption is imposed on either fitted predictor. Define

$$\hat{\mu}_R \triangleq (\hat{\mu}_{R,0}, \hat{\mu}_{R,1}), \quad \hat{\mu}_O \triangleq (\hat{\mu}_{O,0}, \hat{\mu}_{O,1}).$$

Then define the observational treatment-effect prediction

$$g(x) \triangleq \hat{\mu}_{O,1}(x) - \hat{\mu}_{O,0}(x), \quad (4)$$

Definition 2 (RCT-valid AIPW score). Let $q \triangleq (q_0, q_1)$ be a fixed pair of measurable functions from $\mathcal{X}$ to $\mathbb{R}$. Define

$$\varphi(V; q) \triangleq q_1(X) - q_0(X) + \frac{A}{e(X)}\{Y - q_1(X)\} - \frac{1 - A}{1 - e(X)}\{Y - q_0(X)\}. \quad (5)$$

Proposition 1 (RCT-validity for an arbitrary outcome regression). Under Assumptions 1 and 4, every fixed q with finite first moments satisfies

$$\mathbb{E}_R\{\varphi(V; q) \mid X\} = \tau(X). \quad (6)$$

Definition 3 (Fusion score). For $\lambda \in \mathbb{R}$, define

$$\begin{aligned} \hat{\mu}_\lambda &\triangleq (1 - \lambda)\hat{\mu}_R + \lambda\hat{\mu}_O, \\ Z_\lambda &\triangleq \varphi(V; \hat{\mu}_\lambda) = Z_0 + \lambda\Delta, \\ \Delta &\triangleq Z_1 - Z_0. \end{aligned} \quad (7)$$

Equation (6) implies $\mathbb{E}_R(Z_\lambda \mid X) = \tau(X)$ and $\mathbb{E}_R(\Delta \mid X) = 0$. The observational contrast g is used only as a prediction and need not equal τ .

Definition 4 (RCT-only AIPW estimator). Let $\mu_R \triangleq (\mu_{R,0}, \mu_{R,1})$. Setting $q = \mu_R$ in Definition 2 gives the oracle RCT AIPW score. Its nuisance-sample fitted counterpart is $Z_0 = \varphi(V; \hat{\mu}_R)$. Define the RCT-only AIPW estimator on the evaluation sample by

$$\hat{\theta}_{\text{AIPW},R} \triangleq \mathbb{P}_{R,n}\varphi(V; \hat{\mu}_R) = \mathbb{P}_{R,n}Z_0.$$

Corollary 1 (RCT-only AIPW recovery). Under Assumptions 1 and 4, $\hat{\theta}_{\text{AIPW},R}$ is unbiased for θ over D_R^{eval} .

2.3. Existing data-fusion estimators

Three existing strategies provide the nearest comparisons. First, shrinkage methods combine an unbiased experimental estimate with a potentially biased observational estimate and trade variance against observational bias (Rosenman et al., 2023). Second, adaptive RCT-OBS estimators estimate source-specific conditional effects and downweight the observational estimator when its discrepancy from the trial is large (Cheng and Cai, 2021). Third, direct multi-source learners construct weighted pseudo-outcomes for heterogeneous effects across sources (Li et al., 2022). Our estimator differs at the level of the fusion object: λ changes the nuisance regression inside a trial-valid score, while ω centers a fixed prediction across study covariate samples. Neither channel

treats the observational treatment contrast as identified causal truth.

Within the adaptive and integrative line, the elastic procedure of [Yang et al. \(2023\)](#) tests whether real-world data are sufficiently compatible with the randomized trial before pooling information for prespecified effect modifiers. The confounding-function approach of [Yang et al. \(2025\)](#) instead introduces and jointly estimates an HTE function and an observational confounding function under explicit cross-source identification conditions. These methods use observational outcomes through a poolability decision or an identified confounding-function model; our two fusion channels retain the RCT-valid score as the causal anchor.

3. Prediction-Powered Fusion for the ATE

3.1. Estimator and exact variance

Definition 5 (ATE fusion estimator and its components). For fixed (λ, ω) , define

$$\hat{\theta}_r(\lambda, \omega) \triangleq \mathbb{P}_{R,n} Z_\lambda + \omega \{ \mathbb{P}_{O,N} \{ r(X) g(X) \} - \mathbb{P}_{R,n} g(X) \}. \quad (8)$$

- **Target.** $\theta \triangleq \mathbb{E}_R \{ \tau(X) \}$, anchored by the RCT-valid score Z_λ .
- **Nuisances.** Fixed $\hat{\mu}_R, \hat{\mu}_O, g, r$; exact results use $r = r_0$, and g need not be causal.
- **Tuning.** λ blends outcome regressions inside Z_λ , while ω scales the transported control variate.

Holding $q_\lambda, g, \lambda, \omega$ fixed, for a ratio direction h ,

$$D_r \mathbb{E} \{ \hat{\theta}_r(\lambda, \omega) \} [h] = \omega \mathbb{E}_O \{ h(X) g(X) \}. \quad (9)$$

Thus the ratio channel is generally nonorthogonal. In contrast, arbitrary fixed outcome regressions preserve the conditional target under known e_0 , and the g -correction is exactly centered under the exact r_0 . At the coefficient origin,

$$\hat{\theta}_r(0, 0) = \mathbb{P}_{R,n} Z_0 = \hat{\theta}_{\text{AIPW}, R},$$

so the proposed family contains the RCT-only AIPW estimator from Definition 4 exactly. The first term averages an RCT-valid pseudo-outcome. Under Assumption 3, the expression in braces is a mean-zero control variate; under Assumption 2, this is the special case $r = 1$. Thus, the RCT identifies the target, λ changes within-covariate score noise, and ω uses the large OBS covariate sample to remove predictable between-covariate variation.

Theorem 1 (Exact mean and variance under honest evaluation). Under Assumptions 1, 3, and 4, set $r = r_0$, suppose $n, N > 0$, and suppose the displayed scores are square-integrable. Every fixed (λ, ω) then satisfies

$$\mathbb{E} \{ \hat{\theta}_r(\lambda, \omega) \} = \theta, \quad (10)$$

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega)\} = \frac{1}{n} \text{Var}_R\{Z_\lambda - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (11)$$

Equivalently, the law of total variance gives the within-/between-covariate decomposition

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega)\} = \frac{1}{n} \mathbb{E}_R\{\text{Var}_R(Z_\lambda \mid X)\} + \frac{1}{n} \text{Var}_R\{\tau(X) - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (12)$$

The mean follows from $\mathbb{E}_R(Z_\lambda) = \theta$ and $\mathbb{E}_O(rg) = \mathbb{E}_R(g)$. The variance follows after rewriting Equation (8) as the sum of independent RCT and OBS empirical means.

Corollary 2 (Common-marginal specialization). Under Assumptions 1, 2, and 4, set $r = r_0 = 1$ in Theorem 1. Then Equation (8) is exactly unbiased, and its OBS variance term is $\omega^2 \text{Var}_R\{g(X)\}/N$.

3.2. Joint and separable population oracles

Throughout this subsection, Assumption 4 remains in force, n and N are positive, and the displayed random variables have finite second moments.

Definition 6 (ATE variance-quadratic objects). Define the five population moments

$$A \triangleq \text{Var}_R(\Delta), \quad B \triangleq \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg), \quad (13)$$

$$C \triangleq \text{Cov}_R(Z_0, \Delta), \quad D \triangleq \text{Cov}_R(Z_0, g), \quad E \triangleq \text{Cov}_R(\Delta, g). \quad (14)$$

The moment A is the curvature cost of changing the outcome-regression blend λ . The moment B is the curvature cost of the prediction-powered scale ω , including both the RCT subtraction and the scaled OBS addition. The moment C is the baseline-score covariance that determines the linear slope in the λ direction. The moment D is the baseline-prediction covariance that determines the linear slope in the ω direction, with a minus sign because the RCT score contains $-\omega g$. The moment E is the interaction covariance coupling the two coefficients through the cross-term $-2E\lambda\omega$.

Collect these moments and coefficients as

$$\mathbf{H}_{\text{ATE}} \triangleq \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}, \quad \boldsymbol{\ell} \triangleq \begin{pmatrix} C \\ -D \end{pmatrix}, \quad \boldsymbol{\beta} \triangleq \begin{pmatrix} \lambda \\ \omega \end{pmatrix}, \quad (15)$$

Proposition 2 (ATE variance quadratic and positive-semidefinite curvature). Under the standing conditions of this subsection, the compact variance in Equation (11) is

$$\text{Var}\{\hat{\theta}_r(\boldsymbol{\beta})\} = \frac{1}{n} \left\{ \text{Var}_R(Z_0) + 2\boldsymbol{\ell}^\top \boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{H}_{\text{ATE}} \boldsymbol{\beta} \right\}. \quad (16)$$

Moreover, $\mathbf{H}_{\text{ATE}}$ is positive semidefinite because, for every $(u, v) \in \mathbb{R}^2$,

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N}v^2 \text{Var}_O(rg) \geq 0. \quad (17)$$

The variance in Equation (16) contains the RCT-only baseline $\text{Var}_R(Z_0)/n$, which no coefficient can change. Coefficient selection can therefore focus on the remaining quadratic. Its positive-semidefinite curvature guarantees that this reduced objective cannot decrease without bound.

Definition 7 (ATE coefficient objective). Define the coefficient-dependent part of the scaled variance by

$$\begin{aligned} \mathcal{J}_{\text{ATE}}(\beta) &\triangleq 2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta \\ &= 2C\lambda - 2D\omega + A\lambda^2 - 2E\lambda\omega + B\omega^2. \end{aligned}$$

Proposition 3 (Equivalent ATE oracle objectives). For every nonempty feasible set $\Theta \subseteq \mathbb{R}^2$, the two argmin sets, possibly empty when the infimum is not attained, satisfy

$$\arg \min_{\beta \in \Theta} \text{Var}\{\hat{\theta}_r(\beta)\} = \arg \min_{\beta \in \Theta} \mathcal{J}_{\text{ATE}}(\beta).$$

Positive semidefiniteness means that the variance objective cannot curve downward along any direction (u, v) . For this two-dimensional matrix, the stronger condition $AB - E^2 > 0$, together with positive semidefiniteness, makes $\mathbf{H}_{\text{ATE}}$ positive definite and invertible. It turns a possibly flat quadratic into one with a unique unrestricted minimizer.

Definition 8 (Joint and ridge population oracles). When $AB - E^2 > 0$, define the unrestricted joint oracle by

$$\beta_{\text{joint}}^* \triangleq \arg \min_{\beta \in \mathbb{R}^2} \text{Var}\{\hat{\theta}_r(\beta)\} = \arg \min_{\beta \in \mathbb{R}^2} \mathcal{J}_{\text{ATE}}(\beta).$$

The equality of the two argmin sets follows from Proposition 3. For a prespecified $\rho > 0$, define the ridge-stabilized population oracle by

$$\beta_\rho^* \triangleq \arg \min_{\beta \in \mathbb{R}^2} \left\{ \mathcal{J}_{\text{ATE}}(\beta) + \rho \|\beta\|_2^2 \right\}.$$

Let I_2 denote the 2×2 identity matrix.

Theorem 2 (Joint and ridge population oracles). If $AB - E^2 > 0$, the joint oracle in Definition 8 is unique and

$$\beta_{\text{joint}}^* = \begin{pmatrix} \lambda_{\text{joint}}^* \\ \omega_{\text{joint}}^* \end{pmatrix} = -\mathbf{H}_{\text{ATE}}^{-1} \ell = \begin{pmatrix} (ED - BC)/(AB - E^2) \\ (AD - EC)/(AB - E^2) \end{pmatrix}. \quad (18)$$

For every $\rho > 0$, whether or not $\mathbf{H}_{\text{ATE}}$ is singular, the ridge oracle is unique and

$$\beta_\rho^* = -(\mathbf{H}_{\text{ATE}} + \rho I_2)^{-1} \ell. \quad (19)$$

Theorem 2 locates the joint minimum of the two-coefficient variance quadratic. Corollary 2.1 answers the distinct comparison question: it evaluates the quadratic at that minimum and quantifies the improvement relative to the RCT-only origin $\beta = 0$. The general interaction E is retained here because its empirical analogue $\hat{E}$ need not vanish, even though the exact population model later implies $E = 0$.

Corollary 2.1 (Joint-oracle variance reduction). Under the conditions of Theorem 2, let $\sigma_0^2 = \text{Var}_R(Z_0)$ and $\mathcal{V}_0 = \sigma_0^2/n$ be the variance at $(\lambda, \omega) = (0, 0)$. The unrestricted joint oracle has

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \frac{BC^2 - 2ECD + AD^2}{AB - E^2},$$

$$\mathcal{V}_0 - V_{\min} = \frac{1}{n} \ell^\top \mathbf{H}_{\text{ATE}}^{-1} \ell.$$

The reduction is nonnegative and is strictly positive exactly when $\ell \neq 0$, equivalently when $(C, D) \neq (0, 0)$.

Corollary 2.2 (Separable population oracle). Under Assumptions 1, 3, and 4, and the conditions of Theorem 2, suppose $A, B > 0$. The separable population oracle coefficients are

$$\lambda^* \triangleq -\frac{C}{A}, \quad \omega^* \triangleq \frac{D}{B}. \quad (20)$$

The resulting minimum variance is

$$V_{\min}^{\text{sep}} \triangleq \mathcal{V}_0 - \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right). \quad (21)$$

Remark 1 (Singular or constrained curvature). If $\mathbf{H}_{\text{ATE}}$ is singular, an unpenalized joint minimum can lie along a flat set rather than at a unique point. The Moore–Penrose and null-space characterization, as well as existence and uniqueness results under coefficient constraints, are recorded in Appendix B. The ridge oracle remains unique for every $\rho > 0$. In implementation we minimize the continuous quadratic over a prespecified compact rectangle, which guarantees at least one constrained solution.

Under the common covariate marginal,

$$B = \left(1 + \frac{n}{N} \right) \text{Var}_R\{g(X)\}.$$

Define the coefficient rectangle by

$$\Theta_{\text{rect}} \triangleq [L_\lambda, U_\lambda] \times [L_\omega, U_\omega]. \quad (22)$$

Definition 9 (Projection onto an interval). For a closed interval $[L, U]$ with $L \leq U$, define

$$\text{Proj}_{[L, U]}(u) \triangleq \min\{U, \max(L, u)\}. \quad (23)$$

This operator returns u when it lies in the interval and otherwise returns the nearest endpoint.

For positive A, B , the separated objective gives the coordinatewise projection

$$\lambda_{\Theta_{\text{rect}}}^* = \text{Proj}_{[L_\lambda, U_\lambda]}(-C/A), \quad \omega_{\Theta_{\text{rect}}}^* = \text{Proj}_{[L_\omega, U_\omega]}(D/B).$$

For a coupled constraint or a nonzero interaction, the original quadratic must be minimized directly. The population oracle weakly improves the variance objective whenever $(0, 0)$ is feasible. Estimated coefficients have no universal finite-sample no-harm guarantee.

3.3. Estimable honest algorithm

3.3.1. Coefficient estimation and honest evaluation

For $S \in \{R, O\}$, let $m_S \triangleq n_S^{\text{tune}} \geq 2$. For any random quantity H evaluated on D_S^{tune} , define its sample mean by

$$\bar{H}_S \triangleq \frac{1}{m_S} \sum_{i=1}^{m_S} H_{S,i}.$$

For random quantities U and W , conditional on the fixed nuisance fits, define their sample covariance and variance by

$$\begin{aligned} \widehat{\text{Cov}}_S(U, W) &\triangleq \frac{1}{m_S - 1} \sum_{i=1}^{m_S} \{U_{S,i} - \bar{U}_S\} \{W_{S,i} - \bar{W}_S\}, \\ \widehat{\text{Var}}_S(U) &\triangleq \widehat{\text{Cov}}_S(U, U). \end{aligned}$$

The factor n/N below uses the evaluation-sample sizes whose variance is optimized.

Using the tuning samples, compute

$$\hat{A} \triangleq \widehat{\text{Var}}_R(\Delta), \quad \hat{B} \triangleq \widehat{\text{Var}}_R(g) + \frac{n}{N} \widehat{\text{Var}}_O(rg), \quad (24)$$

$$\hat{C} \triangleq \widehat{\text{Cov}}_R(Z_0, \Delta), \quad \hat{D} \triangleq \widehat{\text{Cov}}_R(Z_0, g), \quad \hat{E} \triangleq \widehat{\text{Cov}}_R(\Delta, g). \quad (25)$$

Define

$$\hat{\mathbf{H}}_{\text{ATE}} \triangleq \begin{pmatrix} \hat{A} & -\hat{E} \\ -\hat{E} & \hat{B} \end{pmatrix}, \quad \hat{\ell} \triangleq \begin{pmatrix} \hat{C} \\ -\hat{D} \end{pmatrix}, \quad \hat{d} \triangleq \hat{A}\hat{B} - \hat{E}^2.$$

Before observing the tuning outcomes, prespecify the bounds defining the nonempty compact rectangle Θ_{rect} , require it to contain the origin, and choose positive thresholds t_A, t_B, t_{det} .

Recall that $\hat{\mu}_R = (\hat{\mu}_{R,0}, \hat{\mu}_{R,1})$ and $\hat{\mu}_O = (\hat{\mu}_{O,0}, \hat{\mu}_{O,1})$ are outcome-regression predictors fitted on the RCT and OBS nuisance samples, respectively, and $g \triangleq \hat{\mu}_{O,1} - \hat{\mu}_{O,0}$. No correctness assumption is

imposed on these fitted regressions, and g is not assumed to be a causal effect. Algorithm 1 states the independent-sample implementation covered by the exact finite-sample results. The nuisance, tuning, and evaluation samples may be supplied directly or obtained by an honest prespecified split.

Algorithm 1: ATE estimation with honest evaluation

Input: Nuisance samples $D_R^{\text{nuis}}, D_O^{\text{nuis}}$, tuning samples $D_R^{\text{tune}}, D_O^{\text{tune}}$, evaluation samples $D_R^{\text{eval}}, D_O^{\text{eval}}$, the rectangle Θ_{rect} , and denominator thresholds.

- 1 Fit $\hat{\mu}_R, \hat{\mu}_O, g$, and any ratio model on $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. The exact theorem uses the true r ; replacing it by $\hat{r}$ invokes the drift conditions below.
- 2 Compute the five moments on $D_R^{\text{tune}}, D_O^{\text{tune}}$, and select the coefficients using the axis, unrestricted joint, or constrained quadratic rule.
- 3 Evaluate Equation (8) on $D_R^{\text{eval}}, D_O^{\text{eval}}$. Set $\psi_R \triangleq Z_{\hat{\lambda}} - \hat{\omega}g$ and $\psi_O \triangleq \hat{\omega}rg$, and report $\hat{V} \triangleq \widehat{\text{Var}}_{R,\text{eval}}(\psi_R)/n + \widehat{\text{Var}}_{O,\text{eval}}(\psi_O)/N$. Under the regularity conditions stated in Appendix B, report the normal-approximation confidence interval $\hat{\theta}_r \pm z_{1-\alpha/2}\sqrt{\hat{V}}$.

Output: $\hat{\theta}_r, \hat{V}$, and its normal-approximation confidence interval.

How Algorithm 1 selects the coefficients. The axis rule imposes the population restriction $E = 0$:

$$\hat{\lambda}_{\text{axis}} \triangleq \begin{cases} -\hat{C}/\hat{A}, & \hat{A} > t_A, \\ 0, & \hat{A} \leq t_A, \end{cases} \quad \hat{\omega}_{\text{axis}} \triangleq \begin{cases} \hat{D}/\hat{B}, & \hat{B} > t_B, \\ 0, & \hat{B} \leq t_B. \end{cases}$$

When $\hat{A} > t_A, \hat{B} > t_B$, and $\hat{d} > t_{\text{det}}$, the unrestricted empirical joint rule is

$$\hat{\lambda}_{\text{joint}} \triangleq \frac{\hat{E}\hat{D} - \hat{B}\hat{C}}{\hat{d}}, \quad \hat{\omega}_{\text{joint}} \triangleq \frac{\hat{A}\hat{D} - \hat{E}\hat{C}}{\hat{d}}.$$

The boundary-correct empirical rule is

$$\hat{\beta}_{\Theta_{\text{rect}}} \in \arg \min_{\beta \in \Theta_{\text{rect}}} \{2\hat{\ell}^\top \beta + \beta^\top \hat{\mathbf{H}}_{\text{ATE}} \beta\}.$$

If the unrestricted joint rule lies outside Θ_{rect} , this quadratic program must be solved rather than clipping its coordinates separately. Coordinatewise clipping is exact only after imposing $\hat{E} = 0$. If a required threshold fails or the quadratic solver is numerically unstable, the prespecified joint-rule fallback is $(0, 0)$; the axis rule uses its coordinate-specific threshold definitions above.

Analysis. The restriction-based plug-in imposes $\hat{E} = 0$ and uses $(-\hat{C}/\hat{A}, \hat{D}/\hat{B})$. The empirical joint minimizer retains $\hat{E}$ and applies Equation (18). The two rules differ because the realized tuning covariance $\hat{E}$ is generally nonzero.

Each sample covariance is conditionally unbiased for its corresponding fixed-score population moment given the nuisance fits, but the ratios defining $\hat{\lambda}$ and $\hat{\omega}$ are nonlinear and are not generally unbiased for the population oracle coefficients. The thresholds and feasible rectangle control instability; they do not turn these selected coefficients into finite-sample oracle estimates.

The displayed analytic ratios apply only when their corresponding denominator thresholds pass. Imposing $\hat{E} = 0$ applies the proven population restriction before solving the sample problem; it

does not assert that the realized tuning covariance vanishes. Retaining $\widehat{E} = \widehat{\text{Cov}}_R(\Delta, g)$ instead minimizes the realized empirical quadratic, whose interaction term is generally nonzero because of tuning-sample noise.

For explicit evaluation notation, define

$$\psi_{R,i} \triangleq Z_{\widehat{\lambda}}(V_{R,i}^{\text{eval}}) - \widehat{\omega}g(X_{R,i}^{\text{eval}}), \quad \psi_{O,j} \triangleq \widehat{\omega}r(X_{O,j}^{\text{eval}})g(X_{O,j}^{\text{eval}}),$$

and let $\widehat{\text{Var}}_{R,\text{eval}}$ and $\widehat{\text{Var}}_{O,\text{eval}}$ be the usual unbiased sample variances with denominators $n - 1$ and $N - 1$, respectively. For $n, N \geq 2$, the fixed-selected-coefficient variance estimator and normal-approximation confidence interval are

$$\widehat{V} \triangleq \frac{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)}{n} + \frac{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)}{N},$$

$$\widehat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\widehat{V}}.$$

Unbiasedness of $\widehat{V}$ follows from Assumption 4. Confidence-interval coverage additionally requires the regularity conditions stated precisely in Appendix B. In plain terms, the two evaluation scores must have sufficiently light tails, the RCT and OBS sample sizes must grow at a stable relative rate, and the limiting variance must not collapse to zero. This is an asymptotic, not a finite-sample, coverage statement. Here $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution; for $\alpha = 0.05$, it is approximately 1.96.

Proposition 4 (Exact variance estimation after honest selection). Under Theorem 1, let $(\widehat{\lambda}, \widehat{\omega})$ be selected without the evaluation samples, as required by Assumption 4, and suppose $n, N \geq 2$. The selected estimator is unbiased and

$$\mathbb{E}(\widehat{V}) = \text{Var}\{\widehat{\theta}_r(\widehat{\lambda}, \widehat{\omega})\}.$$

Theorem 3 (Normal approximation after honest evaluation). In addition to Proposition 4, suppose the evaluation regularity conditions stated in Appendix B.12 hold. Then

$$\frac{\widehat{\theta}_r - \theta}{\sqrt{\widehat{V}}} \rightsquigarrow N(0, 1),$$

and the displayed normal-approximation interval has asymptotic coverage $1 - \alpha$. Variance unbiasedness alone does not imply this coverage statement.

Theorem 4 (Exact-ratio oracle variance comparison). Under Assumptions 1, 4, and either Assumption 2 with $r = r_0 = 1$, or Assumption 3 with the exact ratio $r = r_0$, suppose the scores in Equation (11) have finite second moments and $A, B > 0$. Define the scaled conditional variance

$$\widetilde{V}(\lambda, \omega) \triangleq n \text{Var}\{\widehat{\theta}_r(\lambda, \omega)\}, \quad \sigma_0^2 \triangleq \text{Var}_R(Z_0).$$

Here “PPI++-only” means only the member of our estimator family that fixes $\lambda = 0$ and optimally selects the PPI++-style control-variate scale ω . It is not a claim of equivalence to

every PPI++ procedure. The three population-oracle variances are

$$\begin{aligned}\tilde{\mathcal{V}}_{\text{RCT}} &\triangleq \tilde{\mathcal{V}}(0, 0) = \sigma_0^2, \\ \tilde{\mathcal{V}}_{\text{PPI++}} &\triangleq \tilde{\mathcal{V}}(0, D/B) = \sigma_0^2 - \frac{D^2}{B}, \\ \tilde{\mathcal{V}}_{\text{joint}} &\triangleq \tilde{\mathcal{V}}(-C/A, D/B) = \sigma_0^2 - \frac{D^2}{B} - \frac{C^2}{A}.\end{aligned}\tag{26}$$

Consequently,

$$\tilde{\mathcal{V}}_{\text{joint}} \leq \tilde{\mathcal{V}}_{\text{PPI++}} \leq \tilde{\mathcal{V}}_{\text{RCT}}.$$

The second inequality is strict exactly when $D \neq 0$, and the first is strict exactly when $C \neq 0$.

3.3.2. Estimating the density ratio

All ratio estimators below use only the nuisance samples and are fixed before coefficient tuning and final evaluation. Let $\mathbb{P}_R^{\text{nuis}}$ and $\mathbb{P}_O^{\text{nuis}}$ denote their empirical averages.

This subsection follows the path from the target ratio r_0 , through a learned ratio $\hat{r}$, to the tuning moment $\hat{B}_{\hat{r}}$, the selected coefficient $\hat{\omega}_{\hat{r}}$, and the final ATE estimator.

Proposition 5 (Transport and Riesz characterizations). Under either Assumption 2 or Assumption 3, every $h \in L_1(P_R^X)$ satisfies

$$\mathbb{E}_O\{r_0(X)h(X)\} = \mathbb{E}_R\{h(X)\}.\tag{27}$$

Conversely, if $\tilde{r} \geq 0$, $\mathbb{E}_O(\tilde{r}) = 1$, and Equation (27) holds with $\tilde{r}$ in place of r_0 for every bounded measurable h , then $\tilde{r} = r_0$ P_O^X -almost surely. If $r_0 \in L_2(P_O^X)$, it is also the unique minimizer over $L_2(P_O^X)$ of

$$\mathcal{R}(a) \triangleq \mathbb{E}_O\{a(X)^2\} - 2\mathbb{E}_R\{a(X)\}.\tag{28}$$

Equation (27) motivates balance, whereas Equation (28) motivates direct representer learning.

Definition 10 (Classifier density-ratio estimator). The classifier route implements the following steps.

1. Pool the nuisance-sample covariates and label their source by $S = 1$ for RCT and $S = 0$ for OBS.
2. Fit a classifier that estimates $P(S = 1 \mid X = x)$.
3. Correct the fitted odds by the RCT-to-OBS sampling prior.
4. Use the corrected odds as the density-ratio estimate.

Formally, define

$$\pi \triangleq P(S = 1), \quad s_0(x) \triangleq P(S = 1 \mid X = x).$$

When $0 < \pi < 1$, Bayes' rule gives

$$r_0(x) = \frac{1 - \pi}{\pi} \frac{s_0(x)}{1 - s_0(x)}. \quad (29)$$

For a fitted domain probability $\hat{s}$, define

$$\hat{r}_{\text{CLS}}(x) \triangleq \frac{1 - \hat{\pi}}{\hat{\pi}} \frac{\hat{s}(x)}{1 - \hat{s}(x)}. \quad (30)$$

For an unweighted pooled classifier, $\hat{\pi} \triangleq n_R^{\text{nuis}} / (n_R^{\text{nuis}} + n_O^{\text{nuis}})$. Class weighting or resampling changes the effective training prior, so the prior-odds factor must be corrected accordingly. The odds require $0 < \hat{s}(x) < 1$ on the region where the ratio is used. Probability clipping can stabilize the estimate, but it changes the exact density-ratio target.

The classifier route learns the ratio indirectly through source prediction. The following equally available route learns the Riesz representer directly.

Definition 11 (Empirical Riesz ratio learner). For a prespecified function class $\mathcal{A}$, penalty Pen , and $\rho \geq 0$, use a prespecified deterministic rule to break ties among minimizers. Whenever the displayed argmin set is nonempty, choose

$$\hat{r}_{\text{RZ}} \in \arg \min_{a \in \mathcal{A}} \left[\mathbb{P}_O^{\text{nuis}}(a^2) - 2\mathbb{P}_R^{\text{nuis}}(a) + \rho \text{Pen}(a) \right]. \quad (31)$$

Existence can be ensured, for example, by a compact or coercive class, bounded outputs, or sufficient regularization; Lemma 1 states separate existence and uniqueness conditions for the linear-ridge case. This is Riesz-representer learning of the type used in Auto-DML, not itself an Auto-DML estimator (Chernozhukov et al., 2021, 2022). The classifier and Riesz routes are alternative base learners; we do not average them.

Lemma 1 (Linear Riesz solution). Let $a_\gamma(x) \triangleq \phi(x)^\top \gamma$, where $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$, and let $\text{Pen}(a_\gamma) \triangleq \|\gamma\|_2^2$. If $\rho > 0$, or if the empirical Gram matrix is nonsingular when $\rho = 0$, then

$$\hat{\gamma}_\rho \triangleq \left[\mathbb{P}_O^{\text{nuis}}\{\phi(X)\phi(X)^\top\} + \rho I_p \right]^{-1} \mathbb{P}_R^{\text{nuis}}\{\phi(X)\}. \quad (32)$$

Additive splines give a finite linear example, and a frozen neural representation can provide the coordinates of ϕ . End-to-end neural optimization of Equation (31) is generally nonconvex. An unconstrained Riesz fit need not be positive; positivity may be imposed by a positive parameterization, followed by empirical normalization.

Remark 2 (ATE-targeted Riesz loss). Recall that g is the OBS treatment-effect predictor defined in Section 2. For a general ratio candidate a , define

$$\delta_a(X) \triangleq \{a(X) - r_0(X)\}g(X), \quad \delta_{\hat{r}}(X) \triangleq \{\hat{r}(X) - r_0(X)\}g(X). \quad (33)$$

The corresponding population identity is

$$P_O(g^2 a^2) - 2P_R(g^2 a) = \|\delta_a\|_{L_2(P_O)}^2 - P_O(g^2 r_0^2). \quad (34)$$

Its empirical plug-in criterion is

$$\mathbb{P}_O^{\text{nuis}}(g^2 a^2) - 2\mathbb{P}_R^{\text{nuis}}(g^2 a) + \rho \text{Pen}(a). \quad (35)$$

Equation (34) identifies the relevant population error, but consistency of the empirical criterion requires an empirical-process argument or further sample separation.

Either route may now be used as the base estimator. The next definition is an optional common calibration step, not a third way of averaging the two routes.

Definition 12 (Calibrated automatic-balancing ratio). Let $\hat{r}_{\text{base}} > 0$ be a positive ratio obtained from either the classifier or Riesz route, without averaging the two. Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be a prespecified feature map with no intercept coordinate. Use a prespecified deterministic rule to break ties among minimizers. Whenever the displayed argmin set is nonempty, choose

$$\hat{\xi} \in \arg \min_{\xi \in \mathbb{R}^p} \left[\log \mathbb{P}_O^{\text{nuis}} \left\{ \hat{r}_{\text{base}}(X) e^{\xi^\top \phi(X)} \right\} - \xi^\top \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \} \right], \quad (36)$$

$$\hat{r}_{\text{BAL}}(x) \triangleq \frac{\hat{r}_{\text{base}}(x) e^{\hat{\xi}^\top \phi(x)}}{\mathbb{P}_O^{\text{nuis}} \left\{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \right\}}. \quad (37)$$

The denominator in Equation (37) already balances the constant, so an intercept in ϕ would create a nonidentified direction. The minimal ATE choice is the scalar $\phi_{\text{ATE}}(X) \triangleq g(X)$. A richer but finite choice can include g^2 , the fitted outcome regressions, and a small additive-spline or frozen-neural basis. Its dimension p should be small relative to the RCT nuisance-sample size. This exponential tilt follows the calibration principle behind entropy balancing while allowing either positive base learner (Hainmueller, 2012).

Proposition 6 (Automatic balance). If the convex objective in Definition 12 has an interior minimizer, then $\hat{r}_{\text{BAL}}(x) > 0$, $\mathbb{P}_O^{\text{nuis}}(\hat{r}_{\text{BAL}}) = 1$, and the p moment equations

$$\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{BAL}}(X) \phi(X) \} = \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}. \quad (38)$$

These are finite moment equations, not pointwise balance for every x . They are analogous to covariate-balancing propensity scores, which concern a treatment propensity rather than the present source ratio (Imai and Ratkovic, 2014). An exact solution requires the RCT feature mean to lie in the relative interior of the convex hull of the OBS feature vectors under the positive base weights. When this feasibility condition fails, a prespecified penalized approximate-balance objective is required.

With a finite calibrated ratio in hand, the next lemma shows why ratio quality matters for coefficient selection rather than only for covariate balance.

Lemma 2 (Propagation into coefficient selection). For a candidate ratio $\tilde{r}$, write

$$\widehat{B}_{\tilde{r}} \triangleq \widehat{\text{Var}}_R(g) + \frac{n}{N} \widehat{\text{Var}}_O(\tilde{r}g). \quad (39)$$

When the axis threshold for Equation (39) is inactive, define

$$\widehat{\omega}_{\tilde{r}} \triangleq \frac{\widehat{D}}{\widehat{B}_{\tilde{r}}}. \quad (40)$$

With $\delta_{\widehat{r}}$ from Equation (33),

$$\widehat{B}_{\widehat{r}} - \widehat{B}_{r_0} = \frac{n}{N} \left[2\widehat{\text{Cov}}_O(r_0g, \delta_{\widehat{r}}) + \widehat{\text{Var}}_O(\delta_{\widehat{r}}) \right]. \quad (41)$$

When the axis-rule thresholds are inactive,

$$\widehat{\omega}_{\widehat{r}} - \widehat{\omega}_{r_0} = -\widehat{D} \frac{\widehat{B}_{\widehat{r}} - \widehat{B}_{r_0}}{\widehat{B}_{\widehat{r}} \widehat{B}_{r_0}}. \quad (42)$$

For the joint rule, Equation (41) changes the lower-right entry of $\widehat{\mathbf{H}}_{\text{ATE}}$, so both selected coefficients may change.

The coefficient path in Equations (41) and (42) is only one effect of ratio estimation. The next proposition isolates its additional, direct effect on the final ATE estimator.

Proposition 7 (Exact estimated-ratio decomposition). Let $(\widehat{\lambda}, \widehat{\omega})$ be selected without the evaluation observations. Conditional on the fitted objects and selected coefficients,

$$\widehat{\theta}_{\widehat{r}}(\widehat{\lambda}, \widehat{\omega}) - \theta = \widehat{\theta}_{r_0}(\widehat{\lambda}, \widehat{\omega}) - \theta + \widehat{\omega} \mathbb{P}_{O,N} \delta_{\widehat{r}} \quad (43)$$

$$= \widehat{\theta}_{r_0}(\widehat{\lambda}, \widehat{\omega}) - \theta + \widehat{\omega} P_O \delta_{\widehat{r}} + \widehat{\omega} (\mathbb{P}_{O,N} - P_O) \delta_{\widehat{r}}. \quad (44)$$

Equation (41) describes the coefficient-selection path, whereas Equation (44) separates the direct ATE error into population drift and evaluation fluctuation.

Corollary 3 (Exact mean preservation). Under Proposition 7, the ratio correction has zero conditional mean if

$$P_O \delta_{\widehat{r}} = 0 \iff \mathbb{E}_O\{\widehat{r}(X)g(X)\} = \mathbb{E}_R\{g(X)\}. \quad (45)$$

Honest fitting fixes $\widehat{r}$ during evaluation, but does not by itself imply Equation (45).

Corollary 4 (First-order equivalence with the exact-ratio estimator). Suppose $\widehat{r}$ is fitted only on the nuisance samples and is therefore independent of the tuning and evaluation observations. Assume $n/N \rightarrow \kappa \in [0, \infty)$, every tuning moment computed with $\widehat{r}$ converges in

probability to its exact-ratio counterpart, and the selected coefficients converge in probability to a deterministic exact-ratio pair $(\lambda^\circ, \omega^\circ)$. Also suppose the probability of invoking any denominator or determinant fallback tends to zero, and the evaluation conditions of Theorem 3 hold. If

$$\sqrt{n} \hat{\omega} P_O \delta_{\hat{r}} = o_p(1), \quad (46)$$

$$|\hat{\omega}| \sqrt{\frac{n}{N}} \|\delta_{\hat{r}} - P_O \delta_{\hat{r}}\|_{L_2(P_O)} = o_p(1), \quad (47)$$

then the feasible and exact-ratio estimators are first-order equivalent. Hence the normal limit in Theorem 3 and the first-order variance ordering in Theorem 4 are inherited, provided that, for each feasible rule being compared, its $(\lambda^\circ, \omega^\circ)$ equals the corresponding exact-ratio population-oracle pair in that theorem. This does not imply universal finite-sample dominance.

Remark 3 (The ratio nuisance is not orthogonal). For a fixed direction h , the derivative of the estimator's population mean along $r_0 + th$ at $t = 0$ is $\omega \mathbb{E}_O\{h(X)g(X)\}$, which is not generally zero. Calibration may remove drift along selected feature directions, but it does not make the current estimator Neyman-orthogonal to arbitrary ratio perturbations.

3.3.3. Optional nested cross-fitting

As an optional alternative to independent nuisance, tuning, and evaluation samples, observations can be reused without scoring any observation with a nuisance fit trained on its own outcome. Prespecify paired outer folds in both sources, or randomize the folds independently and hold the realized assignments fixed. Within each outer complement, use inner splits to fit nuisances and select $(\hat{\lambda}_k, \hat{\omega}_k)$; then score only the untouched paired folds. With fixed nonnegative weights a_k summing to one, define

$$\hat{\theta}_k \triangleq \mathbb{P}_{R,k} Z_{\hat{\lambda}_k}^{(-k)} + \hat{\omega}_k \left\{ \mathbb{P}_{O,k}(r g^{(-k)}) - \mathbb{P}_{R,k} g^{(-k)} \right\}, \quad \hat{\theta}_{\text{CF}} \triangleq \sum_k a_k \hat{\theta}_k.$$

Proposition 8 (What outer-fold honesty proves). Under Assumption 1 and either the common covariate marginal or exact r , suppose the outer-fold assignment is prespecified and data-independent, or independently randomized with its realized randomness held fixed. Each $\hat{\theta}_k$ is then conditionally unbiased given its own outer complement, and $\hat{\theta}_{\text{CF}}$ is unconditionally unbiased. Overlapping outer training complements make the fold estimators dependent, so these facts do not justify adding foldwise variances or using a naive pooled normal-approximation interval.

A pooled cross-fitted interval requires an additional stability or influence-function argument, for example convergence of fold-specific RCT and OBS scores to common deterministic limits in their respective L_2 norms, uniform $(2 + \delta)$ moments, and a nondegenerate two-sample limit. The present theory does not assert a general cross-fitted variance theorem without such conditions.

3.3.4. Diagnostics

An implementation should report split assignments, propensity bounds, candidate boxes, all denominator and determinant thresholds, coefficient fallbacks, ratio overlap and clipping, and the selected RCT-only comparator. Near-zero $\hat{A}$, $\hat{B}$, or $\hat{d}$ must trigger the prespecified rule rather than an unstable analytic ratio.

4. Prediction-Powered Fusion Learning for CATE

In this section, we develop prediction-powered learners for estimating the CATE in the RCT population. The RCT identifies the causal target, whereas the potentially confounded OBS supplies a treatment-effect prediction that may be statistically informative but need not be causally valid. We preserve an RCT-valid CATE risk and use the OBS prediction only through a transported, mean-zero loss correction. We develop two constructions: the Doubly Robust Fusion Learner, based on an AIPW pseudo-outcome, and the normalized R-Fusion Learner, based on a normalized R-loss.

4.1. CATE targets and RCT-valid learning losses

Definition 13 (CATE target, candidate, and ideal risk). The target is $\tau(x) \triangleq \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}$. A candidate CATE function is a measurable $\zeta \in \mathcal{Z} \subseteq L^2(P_R^X)$, and its ideal risk is

$$\mathcal{R}(\zeta) \triangleq \mathbb{E}_R\{\zeta(X) - \tau(X)\}^2.$$

Nuisance functions are trained on $D_R^{\text{nuis}}, D_O^{\text{nuis}}$, independently of the observations on which a candidate loss is evaluated. Tuning coefficients are selected on the separate validation samples $D_R^{\text{tune}}, D_O^{\text{tune}}$. Throughout each conditional calculation, the nuisance and tuning samples and their learned objects are held fixed, as required by Assumption 4.

Write $q_R \triangleq \hat{\mu}_R$ and $q_O \triangleq \hat{\mu}_O$ for the RCT and OBS outcome-regression pairs defined in Section 2. The first tuning coefficient blends these nuisances inside an RCT-valid score.

Definition 14 (Fused outcome nuisance and pseudo-outcome). For a fixed λ , define

$$q_\lambda \triangleq (1 - \lambda)q_R + \lambda q_O, \quad Z_\lambda \triangleq \varphi(V; q_\lambda).$$

The coefficient λ controls nuisance fusion; it is not a probability that the OBS regression is causally correct.

Lemma 3 (Conditional validity of the fused pseudo-outcome). Under Assumptions 1 and 4, every fixed λ satisfies

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X).$$

The lemma follows directly from Equation (6): the known RCT propensity makes the augmentation terms correct the conditional mean for any fixed outcome-regression pair. It yields the RCT-only learning identity.

Proposition 9 (RCT pseudo-outcome risk decomposition). Suppose $Z_\lambda \in L^2(P_R)$ and $\zeta \in L^2(P_R^X)$, and define

$$Q_{\text{DRF}}(\lambda) \triangleq \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2.$$

Equivalently, $Q_{\text{DRF}}(\lambda) = \mathbb{E}_R\{\text{Var}_R(Z_\lambda \mid X)\}$. Then

$$\mathbb{E}_R\{Z_\lambda - \zeta(X)\}^2 = Q_{\text{DRF}}(\lambda) + \mathcal{R}(\zeta).$$

Thus the RCT-only loss has the correct population minimizers, while $Q_{\text{DRF}}(\lambda)$ records its integrated pseudo-outcome noise.

The residualized route begins from the same target but has a different curvature. Let $e_0(x) \triangleq P_R(A = 1 \mid X = x)$, which is known by RCT design, and define

$$\begin{aligned} m_R(x) &\triangleq e_0(x)q_{R,1}(x) + \{1 - e_0(x)\}q_{R,0}(x), \\ m_O(x) &\triangleq e_0(x)q_{O,1}(x) + \{1 - e_0(x)\}q_{O,0}(x), \\ m_\lambda(x) &\triangleq (1 - \lambda)m_R(x) + \lambda m_O(x), \\ v_0(x) &\triangleq e_0(x)\{1 - e_0(x)\}. \end{aligned}$$

Strict overlap in Assumption 1 gives $v_0(X) > 0$ almost surely.

Lemma 4 (Raw R-loss curvature). For any fixed measurable m such that $Y - m(X) \in L^2(P_R)$, and for $\zeta \in L^2(P_R^X)$, randomization gives

$$\begin{aligned} \mathbb{E}_R[\{A - e_0(X)\}\{Y - m(X)\} \mid X] &= v_0(X)\tau(X), \\ \mathbb{E}_R[\{A - e_0(X)\}^2 \mid X] &= v_0(X). \end{aligned}$$

Consequently, for

$$\ell_R^{\text{raw}}(\zeta; m) \triangleq [Y - m(X) - \{A - e_0(X)\}\zeta(X)]^2,$$

we have

$$\mathbb{E}_R\{\ell_R^{\text{raw}}(\zeta; m) \mid X\} = C_R^{\text{raw}}(X; m) + v_0(X)\{\zeta(X) - \tau(X)\}^2,$$

where

$$C_R^{\text{raw}}(X; m) \triangleq \mathbb{E}_R[\{Y - m(X)\}^2 \mid X] - v_0(X)\tau(X)^2$$

does not depend on ζ .

Lemma 4 shows that a restricted class is projected using the weight $v_0(X)$, not the unweighted ideal risk $\mathcal{R}(\zeta)$. Dividing by this curvature restores the desired target.

Definition 15 (Normalized R-loss and RF noise floor). For any fixed m , define

$$\ell_R^{\text{norm}}(\zeta; m) \triangleq \frac{[Y - m(X) - \{A - e_0(X)\}\zeta(X)]^2}{v_0(X)},$$

$$Q_{\text{RF}}(\lambda) \triangleq \mathbb{E}_R\{\ell_R^{\text{norm}}(\tau; m_\lambda)\}.$$

Proposition 10 (Normalized R-loss risk decomposition). If the displayed losses are integrable, then

$$\mathbb{E}_R\{\ell_R^{\text{norm}}(\zeta; m_\lambda)\} = Q_{\text{RF}}(\lambda) + \mathcal{R}(\zeta).$$

Hence normalized R-loss and the ideal CATE risk have the same minimizers over every fixed restricted class $\mathcal{Z}$. The two constants $Q_{\text{DRF}}(\lambda)$ and $Q_{\text{RF}}(\lambda)$ are different noise floors and need not select the same λ .

Both RCT-only routes identify the desired projection, but neither directly averages the large OBS covariate sample. Let g denote the fitted OBS treatment-effect prediction in Equation (4) and define $G_\zeta(X) \triangleq \{g(X) - \zeta(X)\}^2$. The following full-fusion objects are retained here so that the subsequent orthogonality and finite-sample results remain self-contained.

The primary *Doubly Robust Fusion (DRF) Learner* uses the RCT-valid AIPW score. Here “doubly robust” names that score construction under known e_0 ; it does not assert joint robustness to a shared OBS nuisance or ratio error.

Definition 16 (Doubly Robust Fusion loss). For fixed ζ , define

$$\hat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r) \triangleq \mathbb{P}_R^{\text{fit}}\{Z_\lambda - \zeta(X)\}^2 + \omega \left[\mathbb{P}_O^{\text{fit}}\{r(X)G_\zeta(X)\} - \mathbb{P}_R^{\text{fit}}\{G_\zeta(X)\} \right]. \quad (48)$$

Assume $Z_\lambda, \zeta \in L^2(P_R)$, $G_\zeta \in L^1(P_R)$, and $r_0 G_\zeta \in L^1(P_O)$. The coefficient ω scales the prediction-powered correction and is not part of G_ζ .

Proposition 11 (Exact DRF target). Under Assumptions 1 and 4, suppose either Assumption 2 holds and $r_0 = 1$, or Assumption 3 holds and $r_0 = dP_R^X/dP_O^X$. Use $r = r_0$. Then

$$\mathbb{E}\{\hat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r_0)\} = Q_{\text{DRF}}(\lambda) + \mathcal{R}(\zeta). \quad (49)$$

Thus the population minimizers over any fixed class $\mathcal{Z}$ are the $L_2(P_R^X)$ projections of τ . Under the common covariate marginal, $r_0 = 1$. The expected loss is flat in ω .

The secondary *normalized R-Fusion (RF) Learner* uses the following curvature-matched residual loss.

Definition 17 (Normalized R-Fusion loss). For a fixed marginal-outcome function m , define

$$\kappa(A, X) \triangleq \frac{\{A - e_0(X)\}^2}{v_0(X)}, \quad \mathbb{E}_R(\kappa \mid X) = 1,$$

$$\ell_R^{\text{norm}}(\zeta; m) \triangleq \frac{\{Y - m(X) - \{A - e_0(X)\}\zeta(X)\}^2}{v_0(X)}.$$

The normalization removes the $v_0(X)$ -weighted projection displayed in Lemma 4. Define

$$\widehat{L}_{\text{RF}}(\zeta; m, \omega, r) \triangleq \mathbb{P}_R^{\text{fit}} \ell_R^{\text{norm}}(\zeta; m) + \omega \left[\mathbb{P}_O^{\text{fit}} \{r(X)G_\zeta(X)\} - \mathbb{P}_R^{\text{fit}} \{\kappa(A, X)G_\zeta(X)\} \right]. \quad (50)$$

Assume $\tau \in L^2(P_R)$ and $\ell_R^{\text{norm}}(\zeta; m) \in L^1(P_R)$.

Theorem 5 (Exact normalized RF target). Under the assumptions of Proposition 11,

$$\mathbb{E}\{\widehat{L}_{\text{RF}}(\zeta; m, \omega, r_0)\} = C_R(m) + \mathcal{R}(\zeta). \quad (51)$$

where

$$C_R(m) \triangleq \mathbb{E}_R \left[\frac{\{Y - m(X)\}^2}{e_0(X)\{1 - e_0(X)\}} - \tau(X)^2 \right]$$

is finite and independent of ζ . In particular, $Q_{\text{RF}}(\lambda) = C_R(m_\lambda)$. Hence the population minimizers over a fixed class are the same unweighted $L_2(P_R^X)$ projections of τ .

4.2. Orthogonality audit

Proposition 12 (DRF and RF orthogonality audit). For fixed v, g, r_0 , define

$$\Psi_{\text{DRF}}(\zeta, q)[v] \triangleq \mathbb{E}_R[v(X)\{\varphi(V; q) - \zeta(X)\}],$$

and define the RF projection score by

$$\Psi_{\text{RF}}(\zeta, m)[v] \triangleq \mathbb{E}_R \left[v(X) \frac{A - e_0(X)}{e_0(X)\{1 - e_0(X)\}} \{Y - m(X) - \{A - e_0(X)\}\zeta(X)\} \right].$$

The DRF derivative in q and the RF derivative in m are zero, holding (g, r_0) fixed, although numerical loss derivatives need not vanish. Holding all other nuisances and $\zeta, g, \lambda, \omega$ fixed, the r -partial derivative for either empirical loss is

$$D_r \mathbb{E}\{\widehat{L}_j\}[h] = \omega \mathbb{E}_O[h(X)\{g(X) - \zeta(X)\}^2]. \quad (52)$$

For an estimated propensity, propensity orthogonality holds only locally at $(\zeta, m, e) = (\tau, m_0, e_0)$, where $m_0(X) \triangleq \mathbb{E}_R(Y | X)$. Ratio perturbations remain nonorthogonal as in Equation (52). This is not joint orthogonality when shared OBS regressions generate both a score nuisance and g . Known RCT propensity is load-bearing for arbitrary fused q_λ .

Define

$$\begin{aligned} m_R(x) &\triangleq e_0(x)q_{R,1}(x) + \{1 - e_0(x)\}q_{R,0}(x), \\ m_O(x) &\triangleq e_0(x)q_{O,1}(x) + \{1 - e_0(x)\}q_{O,0}(x), \end{aligned}$$

$$m_\lambda(x) \triangleq (1 - \lambda)m_R(x) + \lambda m_O(x).$$

Theorem 5 holds for every fixed m_λ , so λ is target-preserving but may change residual noise.

4.3. Final learner-risk coefficients and honest selection

For $j \in \{\text{DRF}, \text{RF}\}$, with other learner choices fixed, let $\Theta_{\text{CATE}} \subseteq \mathbb{R}^2$ be a prespecified nonempty coefficient domain. When the minimum is attained, define the algorithm risk and its fusion oracle by

$$\mathfrak{R}_j(\lambda, \omega) \triangleq \mathbb{E}_{\text{fit}} \mathcal{R}(\hat{\zeta}_{\lambda, \omega}^j), \quad (53)$$

$$(\lambda_j^*, \omega_j^*) \in \arg \min_{(\lambda, \omega) \in \Theta_{\text{CATE}}} \mathfrak{R}_j(\lambda, \omega). \quad (54)$$

There is no universal closed form for this algorithm-induced oracle. ATE variance, pseudo-outcome-noise, and fixed-loss-variance oracles are different objects, as detailed in the Appendix.

For honest selection, let $\mathfrak{A}$ be one prespecified finite set of candidates $a = (j, \lambda, \omega, p, \rho, \text{basis}, \dots)$, including an identical RCT-only candidate $a_R \in \mathfrak{A}$. Define

$$\mathcal{R}_{\text{fit}}(a) \triangleq \mathcal{R}(\hat{t}_a). \quad (55)$$

Use one shared RCT-valid Z_{ref} , independent of validation. For $G_a(X) \triangleq \{g(X) - \hat{t}_a(X)\}^2$, define

$$\hat{\mathcal{R}}_a^{\text{ref}} \triangleq \mathbb{P}_{R, \text{val}} \{Z_{\text{ref}} - \hat{t}_a(X)\}^2 + \omega_a \left[\mathbb{P}_{O, \text{val}} \{r^\dagger G_a\} - \mathbb{P}_{R, \text{val}} G_a \right]. \quad (56)$$

Under Assumptions 1 and 4, use $r^\dagger = r_0 = 1$ under Assumption 2, or $r^\dagger = r_0$ under Assumption 3. Then

$$\mathbb{E}_{\text{val}} \hat{\mathcal{R}}_a^{\text{ref}} = Q_{\text{ref}} + \mathcal{R}_{\text{fit}}(a), \quad (57)$$

where $Q_{\text{ref}} \triangleq \mathbb{E}_R \{Z_{\text{ref}} - \tau(X)\}^2$.

Theorem 6 (Finite-class honest validation). Let $M \triangleq |\mathfrak{A}| < \infty$. Suppose the centered RCT and OBS per-observation scores have range lengths at most B_R and B_O . Define

$$\varepsilon_{\text{val}}(\delta) \triangleq B_R \sqrt{\frac{\log(4M/\delta)}{2n_v}} + B_O \sqrt{\frac{\log(4M/\delta)}{2N_v}}.$$

If $\hat{a} \in \arg \min_{a \in \mathfrak{A}} \hat{\mathcal{R}}_a^{\text{ref}}$, then, with probability at least $1 - \delta$,

$$\mathcal{R}_{\text{fit}}(\hat{a}) \leq \min_{a \in \mathfrak{A}} \mathcal{R}_{\text{fit}}(a) + 2\varepsilon_{\text{val}}(\delta). \quad (58)$$

In particular,

$$\mathcal{R}_{\text{fit}}(\hat{a}) \leq \mathcal{R}_{\text{fit}}(a_R) + 2\varepsilon_{\text{val}}(\delta). \quad (59)$$

This is not realized-sample universal dominance.

For an evaluation-independent approximate ratio, define $d_{r,a} \triangleq \omega_a \mathbb{E}_O[(r^\dagger - r_0)G_a]$. If the reference conditional bias is $b_{\text{ref}}(X) \triangleq \mathbb{E}_R(Z_{\text{ref}} - \tau(X) \mid X)$, define $d_{\text{nuis},a} \triangleq 2\mathbb{E}_R[b_{\text{ref}}\{\tau - \hat{t}_a\}]$. Write the corresponding uniform validation drifts as

$$\Delta_r \triangleq \sup_{a \in \mathfrak{A}} |d_{r,a}|, \quad \Delta_{\text{nuis}} \triangleq \sup_{a \in \mathfrak{A}} |d_{\text{nuis},a}|. \quad (60)$$

Theorem 6 then adds $2\Delta_r + 2\Delta_{\text{nuis}}$, provided its concentration envelope holds for the actual weighted scores.

4.4. Why general error analysis is hard and the linear-sieve solution

In general, the empirical inverse Hessian and score depend on the same fitting sample, final-risk coefficients have no universal closed form, and ratio error perturbs both score and curvature. We therefore begin with a fixed linear sieve. Let $b(x) \in \mathbb{R}^p$ be fixed or honestly trained and define $t_\beta(x) \triangleq b(x)^\top \beta$. Let $P \succeq 0$ and $\rho \geq 0$. For DRF, define

$$\begin{aligned} G_R &\triangleq \mathbb{P}_R^{\text{fit}}(bb^\top), & G_{O,r} &\triangleq \mathbb{P}_O^{\text{fit}}(r_0 bb^\top), \\ h_\lambda &\triangleq \mathbb{P}_R^{\text{fit}}(bZ_\lambda), & d_g &\triangleq \mathbb{P}_O^{\text{fit}}(r_0 bg) - \mathbb{P}_R^{\text{fit}}(bg). \end{aligned}$$

Whenever the matrix is invertible,

$$\hat{\beta}_{\lambda,\omega}^{\text{DRF}} = \{(1 - \omega)G_R + \omega G_{O,r} + \rho P\}^{-1} \{h_\lambda + \omega d_g\}. \quad (61)$$

For RF, define

$$\begin{aligned} s_{m_\lambda} &\triangleq \frac{(A - e_0)(Y - m_\lambda)}{e_0(1 - e_0)}, & G_{R,\kappa} &\triangleq \mathbb{P}_R^{\text{fit}}(\kappa bb^\top), \\ h_{m_\lambda} &\triangleq \mathbb{P}_R^{\text{fit}}(bs_{m_\lambda}), & d_{g,\kappa} &\triangleq \mathbb{P}_O^{\text{fit}}(r_0 bg) - \mathbb{P}_R^{\text{fit}}(\kappa bg). \end{aligned}$$

Then

$$\hat{\beta}_{\lambda,\omega}^{\text{RF}} = \{(1 - \omega)G_{R,\kappa} + \omega G_{O,r} + \rho P\}^{-1} \{h_{m_\lambda} + \omega d_{g,\kappa}\}. \quad (62)$$

For $0 \leq \omega \leq 1$, nonnegative r_0 , and $P \succeq 0$, both Hessians are positive semidefinite. A feasible DRF or RF fit using $\hat{r}$ requires $\hat{r} \geq 0$, a direct Hessian check, or an explicit projection onto the positive-semidefinite cone.

For either learner, define

$$\begin{aligned} G &\triangleq \mathbb{E}_R(bb^\top), & \gamma &\triangleq \mathbb{E}_R(b\tau), \\ \beta_p &\triangleq G^{-1}\gamma, & t_p &\triangleq b^\top \beta_p, & a_p^2 &\triangleq \mathbb{E}_R(\tau - t_p)^2. \end{aligned}$$

Assume $G \succ 0$. Define the unpenalized Hessian and right-hand-side objects by

$$\hat{H}_{\text{DRF}} \triangleq (1 - \omega)G_R + \omega G_{O,r}, \quad \hat{h}_{\text{DRF}} \triangleq h_\lambda + \omega d_g,$$

$$\widehat{H}_{\text{RF}} \triangleq (1 - \omega)G_{R,\kappa} + \omega G_{O,r},$$

$$\widehat{h}_{\text{RF}} \triangleq h_{m_\lambda} + \omega d_{g,\kappa}.$$

For either j , the penalized normal equation is

$$(\widehat{H}_j + \rho P)\widehat{\beta}_j = \widehat{h}_j.$$

Define

$$\widehat{S}_j \triangleq \widehat{h}_j - \widehat{H}_j \beta_p.$$

Theorem 7 (Exact sieve excess-risk identity). If $\widehat{H}_j + \rho P$ is invertible, then

$$\widehat{\beta}_j - \beta_p = (\widehat{H}_j + \rho P)^{-1} \{\widehat{S}_j - \rho P \beta_p\}, \quad (63)$$

$$\mathbb{E}_R(\widehat{t}_j - \tau)^2 = a_p^2 + (\widehat{\beta}_j - \beta_p)^\top G(\widehat{\beta}_j - \beta_p). \quad (64)$$

No same-sample bias-variance split follows from $\mathbb{E}(\widehat{S}_j) = 0$, because the inverse Hessian and score are dependent.

Theorem 8 (Fixed-candidate finite-sample bound). Assume the exact-target conditions of Proposition 11 and Theorem 5, known e_0 , exact r_0 , and honestly fixed nuisances. Let fitting sizes be n_f, N_f . Let $\psi_{R,j}$ and $\psi_{O,j}$ denote the raw RCT and OBS vector contributions to $\widehat{S}_j$, including their signs and ω -multipliers. Define

$$\xi_{R,j} \triangleq G^{-1/2} \{\psi_{R,j} - \mathbb{E}_R \psi_{R,j}\}, \quad \xi_{O,j} \triangleq G^{-1/2} \{\psi_{O,j} - \mathbb{E}_O \psi_{O,j}\}.$$

The source means obey the exact cancellation $\mathbb{E}_R \psi_{R,j} + \mathbb{E}_O \psi_{O,j} = 0$, equivalently $\mathbb{E}(\widehat{S}_j) = 0$. Suppose every one-dimensional projection of these centered whitened scores is sub-Gaussian with norm at most $\sigma_{R,j}$ and $\sigma_{O,j}$, respectively. On the Gram event

$$\mathcal{E}_H \triangleq \{\|G^{-1/2}(\widehat{H}_j - G)G^{-1/2}\|_{\text{op}} \leq \eta\}, \quad 0 \leq \eta < 1,$$

the following holds with probability at least $1 - \delta_H - \delta_S$, where $\Pr(\mathcal{E}_H^c) \leq \delta_H$:

$$\begin{aligned} \mathbb{E}_R(\widehat{t}_j - \tau)^2 &\leq a_p^2 + \frac{2C_{\text{sg}}^2}{(1 - \eta)^2} \left[\sigma_{R,j} \sqrt{\frac{p + \log(4/\delta_S)}{n_f}} + \sigma_{O,j} \sqrt{\frac{p + \log(4/\delta_S)}{N_f}} \right]^2 \\ &\quad + \frac{2\rho^2}{(1 - \eta)^2} \|G^{-1/2} P \beta_p\|_2^2. \end{aligned} \quad (65)$$

Here $C_{\text{sg}} > 0$ is a universal constant from Euclidean-norm concentration for means of independent sub-Gaussian vectors. The order $a_p^2 + O_p(p/n_f + p/N_f)$ additionally requires bounded score envelopes, a Gram event bounded away from singularity with probability tending to one, and ridge bias no larger than that order.

With an honest estimated ratio, the conditional score and Hessian drifts are

$$d_{r,j} \triangleq \omega \mathbb{E}_O[(\widehat{r} - r_0)b(g - t_p)], \quad (66)$$

$$H_{r,j} \triangleq \omega \mathbb{E}_O[(\widehat{r} - r_0)bb^\top]. \quad (67)$$

Normalization alone controls neither object.

Corollary 5 (Operational additive-spline rate). Suppose $\tau(x) = \sum_{\ell=1}^d \tau_\ell(x_\ell)$, with fixed d and smoothness $s > 0$. Consider a deterministic K_n , or a prespecified finite grid selected by Theorem 6. Let $p_n \asymp K_n$, $a_{p_n}^2 = O(K_n^{-2s})$, and $\delta_n \downarrow 0$. Assume exact r_0 , known e_0 , and uniformly along the sequence: (i) the eigenvalues of G_{p_n} are bounded above and away from zero; (ii) the centered whitened scores and weighted designs are sub-Gaussian with bounded norms, while r_0 and κ are nonnegative and bounded; (iii) $K_n + \log(1/\delta_n) = o\{\min(n_f, N_f)\}$; and (iv)

$$\rho_n^2 \|G_{p_n}^{-1/2} P \beta_{p_n}\|_2^2 = O\left(K_n^{-2s} + \frac{K_n}{n_f} + \frac{K_n}{N_f}\right).$$

If a finite grid is used, also require the validation remainder in Theorem 6 to be no larger than this target rate. Then

$$\mathbb{E}_R(\hat{t}_j - \tau)^2 = O_p\left(K_n^{-2s} + \frac{K_n}{n_f} + \frac{K_n}{N_f}\right). \quad (68)$$

With $m_{\text{eff}} \triangleq (1/n_f + 1/N_f)^{-1}$ and $K_n \asymp m_{\text{eff}}^{1/(2s+1)}$, this becomes $O_p\{m_{\text{eff}}^{-2s/(2s+1)}\}$, provided the same conditions hold. If $N_f \geq cn_f$, the RCT size determines the generic exponent.

4.5. Implementation boundaries and open theory

A fixed or honestly trained frozen neural representation can replace b , conditional on that representation. End-to-end neural fitting is nonconvex and is not covered. Estimated-ratio guarantees require uniform score, Hessian, and validation-loss control. Cross-fitting can recover data efficiency, but overlapping outer training sets do not justify a naive sum of fold variances. Neither target preservation nor candidate inclusion gives universal finite-sample improvement for estimated (λ, ω) .

5. Experiments

5.1. Completed ATE benchmarks

The completed experiments evaluate ATE estimation only. All nested benchmarks use 20 Monte Carlo repetitions per cell, an RCT-only AIPW baseline, a λ -only outcome-model fusion estimator, an ω -only PPI correction estimator, and the joint estimator. The code retains two stale internal names: `haipw_only` denotes the λ -only estimator and `ppi_power_tuned_only` denotes the ω -only estimator. The first label does not identify the method with the separate HAIPW literature.

The fixed-OBS-budget benchmark contains 16 design cells: three arbitrary nonlinear SCMs and one STAR real- X construction, each under a common covariate marginal and covariate shift and $n_R \in \{20, 30\}$, with $N_O = 5000$. The joint estimator has the smallest RMSE in 11 cells, and the λ -only estimator is best in the remaining five. The joint estimator has lower RMSE than AIPW

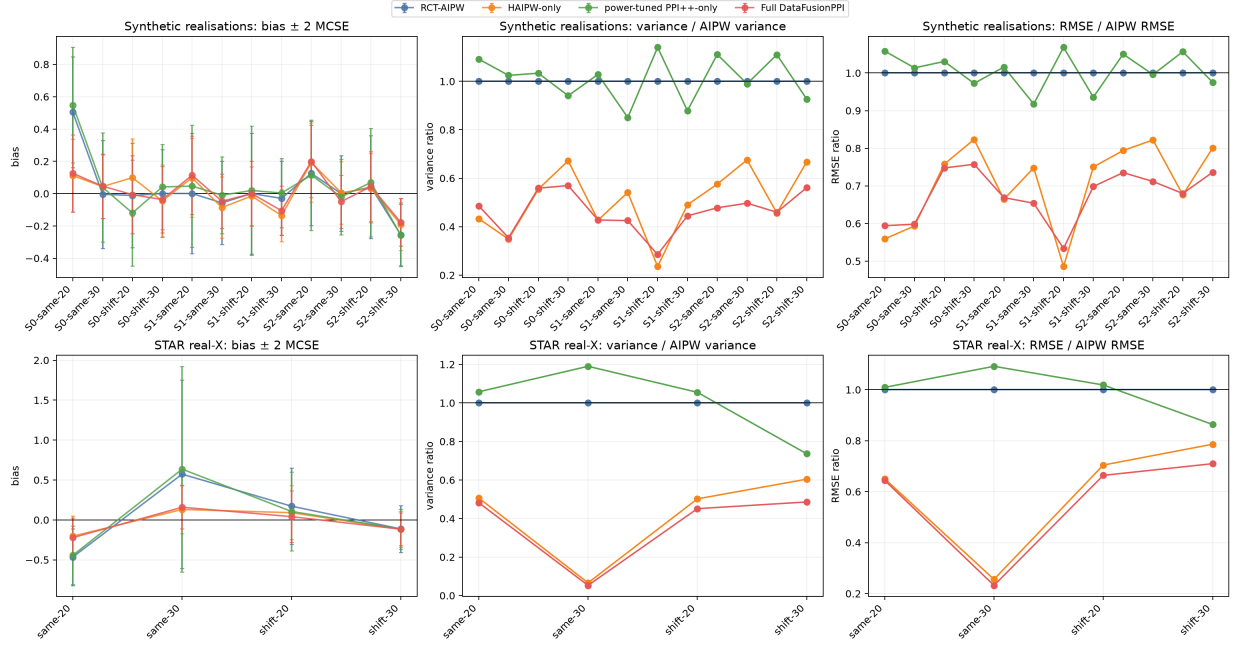


Figure 1: Fixed-OBS-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30\}$ and $N_O = 5000$.

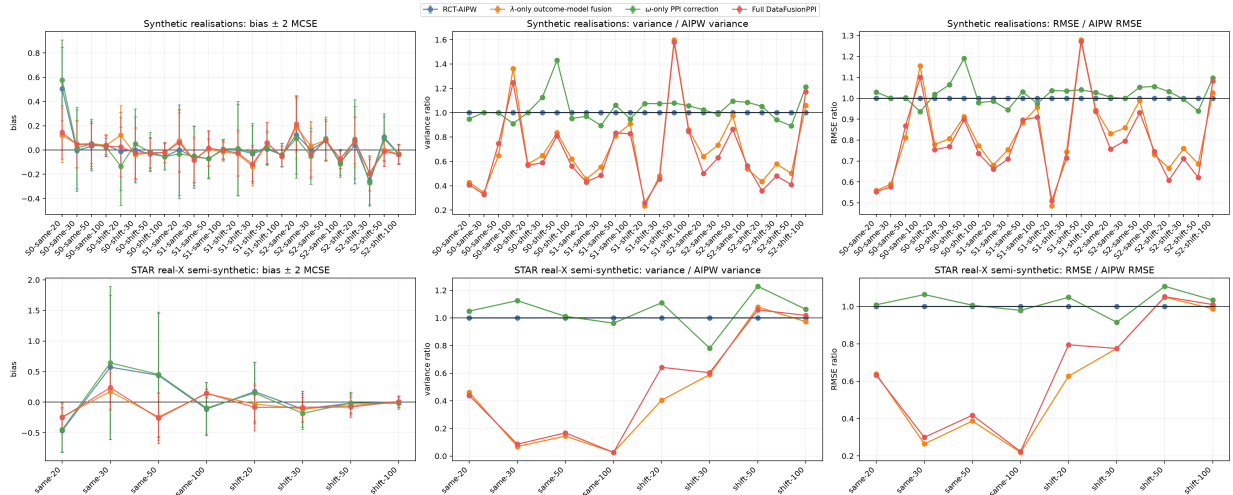


Figure 2: Proportional-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

in all 16 cells, with mean RMSE ratio 0.648 and mean empirical-variance ratio 0.439 relative to AIPW. The λ -only estimator also beats AIPW in all 16 cells, with mean RMSE ratio 0.679. The ω -only estimator is not best in any cell and has mean RMSE ratio 1.005.

The proportional-budget benchmark contains 32 cells with $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$. The joint estimator has the smallest RMSE in 18 cells, the λ -only estimator in 10, the ω -only estimator in one, and AIPW in three. The joint estimator has lower RMSE than AIPW in 27 of 32 cells, with mean RMSE ratio 0.760 and mean empirical-variance ratio 0.625. The λ -only estimator beats AIPW in 28 cells and has mean RMSE ratio 0.769. The ω -only estimator beats AIPW in 10 cells and has mean RMSE ratio 1.020. These counts show that joint tuning is often useful but is not uniformly best after coefficient estimation.

The STAR benchmark uses only real kindergarten covariates from Project STAR. Treatment assignments, latent confounding, potential outcomes, and effect heterogeneity are generated synthetically. It is therefore a real- X , semi-synthetic benchmark rather than an analysis of STAR’s original intervention or outcomes. The planned WHI CT+OS analysis has not been run because participant-level access, a semantic crosswalk, and a survival or censoring extension are still required.

5.2. Planned CATE evaluation and interpretation

CATE experiments are planned. The primary design will compare the proved AIPW-pseudo-outcome F-Learner with the prospective R-loss extension using additive cubic B-splines, frozen neural bases, and linear sieves under a common covariate marginal and covariate shift. Each study will compare trial-only, λ -only, ω -only, and joint tuning using held-out RCT-population CATE risk. Classifier-only, balancing-only, hybrid, and oracle transport weights will be separated. No CATE result is claimed from Figures 1 and 2.

The empirical conclusion is deliberately limited. The two-channel population oracle weakly improves the ATE variance objective when the trial-only coefficient is feasible, but estimated coefficients do not inherit a universal no-harm property. The present 20-repetition studies suggest that the λ channel is the more stable source of gain and that joint tuning can add value in some designs. More Monte Carlo repetitions, completed CATE experiments, additional real- X supports, and an authorized real-outcome analysis are required before making a stronger empirical claim.

5.3. Scope, failure modes, and conclusion

The framework uses observational information through prediction rather than by declaring the observational treatment contrast causal. Trial randomization validates the pseudo-outcome for every fixed outcome regression. A common covariate marginal or an exact density ratio centers the ATE control variate and the prediction-powered CATE loss, which creates separate λ - and ω -channels while retaining RCT-population targets.

The first boundary is covariate transport. Without weighting under $P_R^X \neq P_O^X$, the ATE correction has drift $\omega\{P_O g - P_R g\}$, and the unweighted CATE loss targets the wrong covariate average. Exact $r = dP_R^X/dP_O^X$ removes both discrepancies. An estimated ratio instead contributes the explicit ATE drift $\omega P_O\{(\hat{r} - r)g\}$ and the CATE validation drift in Equation (60). Evaluation-only variance is first-order adequate only under the stated negligible-drift conditions; otherwise ratio-training influence and covariance terms are required.

The second boundary is trial propensity overlap. Proposition 1 uses the known, strictly positive randomized-trial propensity. Extreme inverse-propensity factors can make the required moments unstable even when the conditional-mean identity remains algebraically valid, and an incorrect propensity removes the exact arbitrary- q guarantee.

The third boundary is weak observational prediction. The unrestricted ω -gain depends on how strongly $g(X)$ predicts $\tau(X)$, with attenuation from the finite n/N ratio. The λ -channel gives little gain when $\Delta = Z_1 - Z_0$ is uninformative or nearly degenerate. Population oracles diagnose these limits but do not make their empirical coefficient estimates stable.

The fourth boundary is adaptive reuse. Fitting nuisances, tuning candidates, and evaluating the selected candidate on the same outcomes invalidates the conditioning product law. Independent nuisance, tuning, and evaluation samples prevent direct leakage; they may be supplied separately or constructed by splitting. Optional nested outer cross-fitting preserves foldwise and aggregate unbiasedness under exact transport and fold honesty, but overlapping training complements invalidate a naive sum of fold variances. Cross-fitted inference needs an additional stability or influence-function theorem.

The fifth boundary is learner restriction and curvature. Target preservation does not make a finite-sample restricted or highly penalized projection accurate. The square-completed representation is a nonnegative mixture only for $0 \leq \omega < 1$, and positive semidefiniteness at $\omega = 1$ must be read from the original quadratic. Coefficients outside $[0, 1]$, nonlinear parameterizations, and penalty rescaling require direct analysis of the original objective.

The sixth boundary is tail behavior. ATE variance identities require square-integrable scores; the Wald theorem additionally requires conditional Lindeberg control, a sufficient $(2 + \delta)$ -moment condition, and a nondegenerate variance limit. Fixed-loss variance tuning imposes fourth-moment-type requirements. Heavy tails can therefore destabilize oracle formulas even when their algebraic denominators are nonzero.

Within these boundaries, the randomized trial fixes the causal target, the OBS outcome model can reduce within- X pseudo-outcome noise through λ , and the OBS covariate sample can reduce predictable between- X variation through ω . The exact results are conditional identities and population-oracle comparisons. Empirical work must retain the identical trial-only fallback, report selection and denominator instability, and avoid interpreting them as a universal finite-sample no-harm guarantee.

References

- Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., and Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671):669–674.
- Angelopoulos, A. N., Duchi, J. C., and Zrnic, T. (2024). PPI++: Efficient prediction-powered inference. *Journal of Machine Learning Research*, 25(104):1–59.
- Bareinboim, E. and Pearl, J. (2016). Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352.
- Cheng, D. and Cai, T. (2021). Adaptive combination of randomized and observational data. *Journal of the American Statistical Association*, 116(536):1863–1874.
- Chernozhukov, V., Newey, W. K., and Singh, R. (2021). De-biased machine learning of global and local parameters using regularized riesz representers.
- Chernozhukov, V., Newey, W. K., and Singh, R. (2022). Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Imai, K. and Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B*, 76(1):243–263.
- Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049.
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165.
- Li, X., Li, Y., Cui, Q., Li, L., and Zhou, J. (2022). Robust direct learning for causal data fusion. In *Proceedings of the 14th Asian Conference on Machine Learning*, volume 189 of *Proceedings of Machine Learning Research*.
- Nie, X. and Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319.
- Rosenman, E., Basse, G., Owen, A., and Baiocchi, M. (2023). Combining observational and experimental datasets using shrinkage estimators. *Biometrics*, 79(4):2961–2973.
- Shi, X., Pan, Z., and Miao, W. (2023). Data integration in causal inference. *WIREs Computational Statistics*, 15(1):e1581.
- Yang, S., Gao, C., Zeng, D., and Wang, X. (2023). Elastic integrative analysis of randomised trial and real-world data for treatment heterogeneity estimation. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(3):575–596.
- Yang, S., Liu, S., Zeng, D., and Wang, X. (2025). Data fusion methods for the heterogeneity of treatment effect and confounding function. *Bernoulli*, 31(4):2987–3012.

A. Simulation Details

All completed simulation artifacts were generated with base seed 190602. The primary nested benchmarks use 20 outer Monte Carlo repetitions per design cell and compare four estimators: RCT-only AIPW, λ -only outcome-model fusion, ω -only PPI correction, and joint (λ, ω) fusion. The implementation retains the historical internal codes `haipw_only` and `ppi_power_tuned_only` for the second and third estimators, respectively. These codes are not the user-facing method names, and the first does not refer to the separate HAIPW paper.

The arbitrary simulation family uses three nonlinear structural causal models. In every model, a domain indicator S distinguishes the observational sample ($S = 0$) from the randomized sample ($S = 1$); covariates depend on latent variables, observational treatment depends on both measured covariates and an unmeasured confounder, trial treatment follows the known randomized propensity, and outcomes may depend nonlinearly on covariates, treatment, and the latent confounder. The common-marginal regime uses $P_R^X = P_O^X$. The shifted regime changes the covariate distribution while preserving overlap and estimates the density ratio from a domain classifier. The same estimators and evaluation logic are used in both regimes.

The fixed-OBS-budget experiment has 16 cells: four benchmark families, two covariate regimes, and $n_R \in \{20, 30\}$, with $N_O = 5000$. The proportional experiment has 32 cells: four benchmark families, two regimes, and $n_R \in \{20, 30, 50, 100\}$, with the exact mapping $N_O = 100n_R \in \{2000, 3000, 5000, 10000\}$. Coefficients are estimated by the nested honest procedure; no oracle coefficient or oracle density ratio is used in the reported method comparison. The replication-level CSV files, summaries, and composite plots in `materials/` are the canonical numerical artifacts.

The fourth benchmark family reuses the empirical covariate support from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment. Treatment, latent confounding, outcomes, and treatment-effect heterogeneity are generated by the benchmark code. Consequently this is a real- X , semi-synthetic design, not an evaluation of the original STAR outcomes or intervention effect.

Figure 3 records the earlier two-method pilot. It is retained for provenance but is not the basis for the four-method conclusions in Section 5.

No participant-level WHI CT+OS analysis is included. Such an analysis requires authorized data access, a verified trial-observational semantic crosswalk, a prespecified estimand, and methods appropriate for censoring if the selected outcome is time-to-event. No completed CATE experiment or real-outcome result is represented by the present artifacts.

Illustrative random-SCM debugging pilot (20 outer refits per cell)
per stage: RCT n / OBS 5000; total generated per replicate: $3n / 15000$

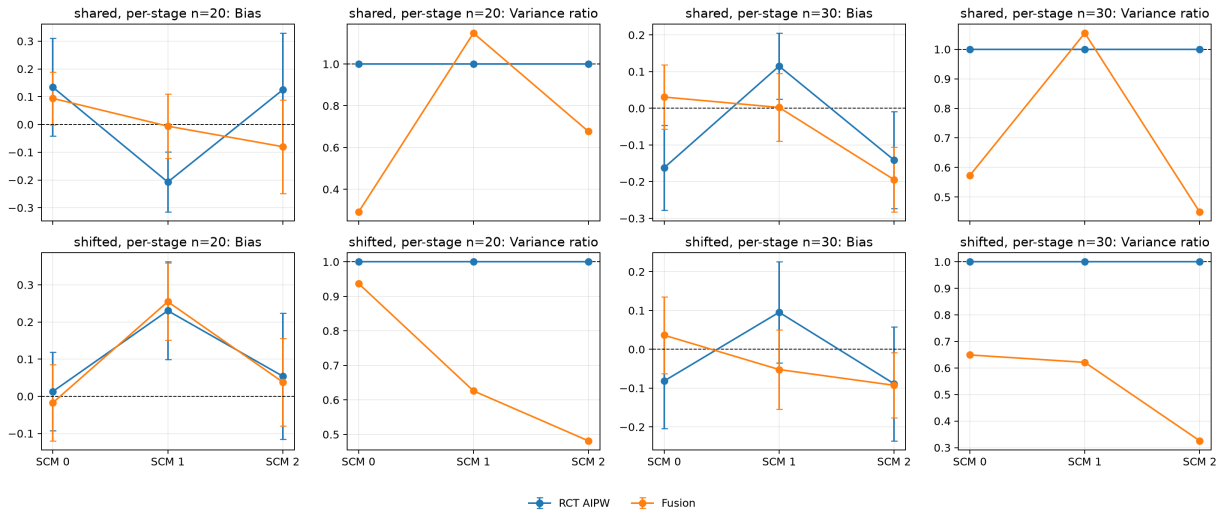


Figure 3: Earlier SSEM ATE pilot. This artifact predates the nested four-method comparison and should be interpreted only as a design and implementation check.

B. Proof Details

Unless stated otherwise, all proofs use Assumption 4: every nuisance, ratio, candidate function, and coefficient is fixed before the independent RCT and OBS evaluation observations are used.

B.1. Conditional validity of the trial pseudo-outcome

Fix $q = (q_0, q_1)$ without using the evaluation samples and condition on X . Under trial randomization, consistency, and honest sample separation,

$$\mathbb{E}_R \left[\frac{A}{e(X)} \{Y - q_1(X)\} \mid X \right] = \mathbb{E}_R \{Y(1) - q_1(X) \mid X\}, \quad (69)$$

$$\mathbb{E}_R \left[\frac{1 - A}{1 - e(X)} \{Y - q_0(X)\} \mid X \right] = \mathbb{E}_R \{Y(0) - q_0(X) \mid X\}. \quad (70)$$

Substituting these two equalities into the definition of φ cancels the two fitted regressions and yields $\mathbb{E}_R \{\varphi(V; q) \mid X\} = \tau(X)$. Applying the identity to $q = \hat{\mu}_R$, $q = \hat{\mu}_O$, and their affine blend gives

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X), \quad \mathbb{E}_R(\Delta \mid X) = 0. \quad (71)$$

B.2. Proof of Corollary 1

Apply Equation (6) with the nuisance-sample fitted pair $q = \hat{\mu}_R$, average over D_R^{eval} , and use iterated expectation. This gives

$$\mathbb{E}\{\hat{\theta}_{\text{AIPW}, R}\} = \mathbb{E}_R\{\tau(X)\} = \theta.$$

B.3. Proof of Theorem 1

Using $\mathbb{E}_R(Z_\lambda) = \theta$ and exact transport,

$$\mathbb{E}_O\{r(X)g(X)\} = \mathbb{E}_R\{g(X)\}, \quad (72)$$

the evaluation-sample mean of Equation (8) is θ . Rewrite the centered estimator as

$$\hat{\theta}_r(\lambda, \omega) - \theta = (\mathbb{P}_{R,n} - P_R)\{Z_\lambda - \omega g\} + (\mathbb{P}_{O,N} - P_O)\{\omega r g\}. \quad (73)$$

The two summands are independent because D_R^{eval} and D_O^{eval} are independent. The variance of an empirical mean of m independent observations is the population variance divided by m , which proves Equation (11). Unbiased sample variances on the two independent evaluation blocks therefore give the displayed fixed-selected-coefficient variance estimator in Section 3. Coverage of its Wald interval additionally uses the stated evaluation-sample Lindeberg and moment assumptions; variance unbiasedness alone is not a central limit theorem.

Because $g(X)$ is fixed given X ,

$$\mathbb{E}_R\{Z_\lambda - \omega g(X) \mid X\} = \tau(X) - \omega g(X), \quad \text{Var}_R\{Z_\lambda - \omega g(X) \mid X\} = \text{Var}_R(Z_\lambda \mid X).$$

Applying the law of total variance proves Equation (12).

B.4. Proof of Proposition 2

Expand the first variance in Equation (11) using $Z_\lambda = Z_0 + \lambda\Delta$. After collecting terms and multiplying by n , the coefficient-dependent part is

$$2C\lambda - 2D\omega + A\lambda^2 - 2E\lambda\omega + B\omega^2, \quad (74)$$

which is $\mathcal{J}_{\text{ATE}}(\beta)$. This proves Equation (16). The representation

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N}v^2 \text{Var}_O(rg) \quad (75)$$

proves positive semidefiniteness.

B.5. Proof of Proposition 3

Equation (16) is the positive affine transformation

$$\text{Var}\{\hat{\theta}_r(\beta)\} = \frac{1}{n} \text{Var}_R(Z_0) + \frac{1}{n} \mathcal{J}_{\text{ATE}}(\beta),$$

where $n > 0$. Adding a constant and multiplying by a positive scalar preserve the complete argmin set over every feasible set.

B.6. Proof of Theorem 2

If $AB - E^2 > 0$, $\mathbf{H}_{\text{ATE}}$ is positive definite. The first-order condition for $\mathcal{J}_{\text{ATE}}$ is $\mathbf{H}_{\text{ATE}}\beta = -\ell$, which has the unique solution

$$\beta_{\text{joint}}^* = -\mathbf{H}_{\text{ATE}}^{-1}\ell = \begin{pmatrix} (ED - BC)/(AB - E^2) \\ (AD - EC)/(AB - E^2) \end{pmatrix}.$$

For $\rho > 0$, the ridge objective has gradient

$$\nabla_\beta \left\{ \mathcal{J}_{\text{ATE}}(\beta) + \rho \|\beta\|_2^2 \right\} = 2\ell + 2(\mathbf{H}_{\text{ATE}} + \rho I_2)\beta.$$

Positive semidefiniteness of $\mathbf{H}_{\text{ATE}}$ makes $\mathbf{H}_{\text{ATE}} + \rho I_2$ positive definite. Setting the gradient to zero proves Equation (19) and uniqueness.

B.7. Singular and constrained extensions

When $\mathbf{H}_{\text{ATE}}$ is singular, nonnegativity of the variance quadratic implies $\ell \in \text{Range}(\mathbf{H}_{\text{ATE}})$; otherwise a null-space direction would make the expression arbitrarily negative. The complete minimizer

set is therefore $-\mathbf{H}_{\text{ATE}}^+ \boldsymbol{\ell} + \text{Null}(\mathbf{H}_{\text{ATE}})$. Continuity of the quadratic proves attainment on every prespecified compact feasible set.

For completeness, suppose $\mathbf{H}_{\text{ATE}}$ is positive definite and write

$$\mathcal{J}_{\text{ATE}}(\boldsymbol{\beta}) = 2\boldsymbol{\ell}^\top \boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{H}_{\text{ATE}} \boldsymbol{\beta}, \quad \boldsymbol{\beta}_{\text{joint}}^* = -\mathbf{H}_{\text{ATE}}^{-1} \boldsymbol{\ell}.$$

Completing the square gives

$$\mathcal{J}_{\text{ATE}}(\boldsymbol{\beta}) - \mathcal{J}_{\text{ATE}}(\boldsymbol{\beta}_{\text{joint}}^*) = (\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{joint}}^*)^\top \mathbf{H}_{\text{ATE}} (\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{joint}}^*).$$

The norm induced by a positive-definite matrix is equivalent to the Euclidean norm. Hence this distance is coercive and strictly convex. On every nonempty closed convex feasible set, it therefore attains a unique minimum, namely the $\mathbf{H}_{\text{ATE}}$ -metric projection of $\boldsymbol{\beta}_{\text{joint}}^*$.

If $A = \text{Var}_R(\Delta) = 0$, then Δ is constant almost surely, and $\mathbb{E}_R(\Delta \mid X) = 0$ forces that constant to be zero. If

$$B = \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg) = 0,$$

then g is constant P_R -almost surely and rg is constant P_O -almost surely. Exact transport gives $P_O(rg) = P_R(g)$, so the two constants agree and $\mathbb{P}_{O,N}(rg) - \mathbb{P}_{R,n}(g) = 0$ for every evaluation sample.

B.8. Proof of Corollary 2.1

Write the evaluation-sample variance as

$$V(\boldsymbol{\beta}) = \mathcal{V}_0 + \frac{1}{n} \{2\boldsymbol{\ell}^\top \boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{H}_{\text{ATE}} \boldsymbol{\beta}\}, \quad \mathcal{V}_0 = \frac{1}{n} \text{Var}_R(Z_0).$$

When $AB - E^2 > 0$, the matrix $\mathbf{H}_{\text{ATE}}$ is positive definite and the unique minimizer is $\boldsymbol{\beta}_{\text{joint}}^* = -\mathbf{H}_{\text{ATE}}^{-1} \boldsymbol{\ell}$. Substitution gives

$$V(\boldsymbol{\beta}_{\text{joint}}^*) = \mathcal{V}_0 - \frac{1}{n} \boldsymbol{\ell}^\top \mathbf{H}_{\text{ATE}}^{-1} \boldsymbol{\ell}.$$

For $\mathbf{H}_{\text{ATE}} = \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}$,

$$\begin{aligned} \mathbf{H}_{\text{ATE}}^{-1} &= \frac{1}{AB - E^2} \begin{pmatrix} B & E \\ E & A \end{pmatrix}, \\ \boldsymbol{\ell}^\top \mathbf{H}_{\text{ATE}}^{-1} \boldsymbol{\ell} &= \frac{BC^2 - 2ECD + AD^2}{AB - E^2}. \end{aligned}$$

Positive definiteness makes this quadratic form nonnegative and makes it zero exactly when $\boldsymbol{\ell} = 0$.

Singular variance reduction and compact constraints. If $\mathbf{H}_{\text{ATE}}$ is singular, boundedness below of the variance objective implies $\boldsymbol{\ell} \in \text{Range}(\mathbf{H}_{\text{ATE}})$. Evaluating at the minimum-norm solution

$-\mathbf{H}_{\text{ATE}}^+ \ell$ gives

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \ell^\top \mathbf{H}_{\text{ATE}}^+ \ell.$$

The pseudoinverse is positive definite on $\text{Range}(\mathbf{H}_{\text{ATE}})$, so the reduction is strictly positive exactly when $\ell \neq 0$. Finally, continuity gives attainment on every nonempty compact feasible set Θ . If $0 \in \Theta$, the constrained minimum is at most $V(0) = \mathcal{V}_0$, and it is strictly smaller exactly when some feasible β makes $2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta < 0$.

B.9. Proof of Corollary 2.2

Equation (6) gives $\mathbb{E}_R(Z_a \mid X) = \tau(X)$ for $a \in \{0, 1\}$. Subtracting the two equalities yields $\mathbb{E}_R(\Delta \mid X) = 0$. Since g is a fixed function of X , iterated expectation gives

$$E = \mathbb{E}_R[\{g(X) - \mathbb{E}_R g(X)\} \mathbb{E}_R(\Delta \mid X)] = 0, \quad (76)$$

$$D = \text{Cov}_R\{\mathbb{E}_R(Z_0 \mid X), g(X)\} = \text{Cov}_R\{\tau(X), g(X)\}. \quad (77)$$

Substituting $E = 0$ into the coefficient objective and completing the two scalar squares gives

$$\mathcal{J}_{\text{ATE}}(\beta) = A \left(\lambda + \frac{C}{A} \right)^2 + B \left(\omega - \frac{D}{B} \right)^2 - \frac{C^2}{A} - \frac{D^2}{B}.$$

The two minimizers are therefore $-C/A$ and D/B , proving Equation (20). Substitution into the variance in Equation (16) gives

$$V_{\min}^{\text{sep}} = \mathcal{V}_0 - \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right),$$

which proves Equation (21).

B.10. Prediction-powered CATE losses and finite-sample theory

Fix $t, \lambda, \omega, g, r_0$ before using the fitting sample. Conditional on X , pseudo-outcome validity gives $\mathbb{E}_R(Z_\lambda - \tau(X) \mid X) = 0$, and hence

$$\mathbb{E}_R\{Z_\lambda - t(X)\}^2 = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2 + \mathbb{E}_R\{t(X) - \tau(X)\}^2. \quad (78)$$

Exact transport also gives

$$\mathbb{E}_O\{r_0(X)(g(X) - t(X))^2\} = \mathbb{E}_R\{g(X) - t(X)\}^2. \quad (79)$$

The correction in Equation (48) therefore has mean zero, which proves Proposition 11.

For Proposition 12, let $q_\varepsilon = q + \varepsilon h$. Direct differentiation gives

$$\dot{Z}_q[h] = -\{A - e_0(X)\} \left\{ \frac{h_1(X)}{e_0(X)} + \frac{h_0(X)}{1 - e_0(X)} \right\}.$$

Its conditional mean is zero, so $\mathbb{E}_R\{v(X)\dot{Z}_q[h]\} = 0$. Differentiating the numerical loss instead gives $2\mathbb{E}_R[(Z(q) - t)\dot{Z}_q[h]]$, which need not vanish. Holding the remaining nuisances and t, g, λ, ω fixed, differentiating the positive OBS term in either expected empirical loss with respect to r proves Equation (52).

For the normalized RF identity, expand conditionally on X . Randomization implies

$$\begin{aligned}\mathbb{E}_R[(A - e_0)\{Y - m(X)\} \mid X] &= e_0(X)\{1 - e_0(X)\}\tau(X), \\ \mathbb{E}_R[\{A - e_0(X)\}^2 \mid X] &= e_0(X)\{1 - e_0(X)\}.\end{aligned}$$

It follows that

$$\mathbb{E}_R \ell_R^{\text{norm}}(t; m) = C_R(m) + \mathbb{E}_R(t - \tau)^2.$$

Because $\mathbb{E}_R(\kappa \mid X) = 1$ and exact transport holds, the RF correction has mean zero. This proves Theorem 5. For the RF part of Proposition 12, differentiation with respect to m gives a multiple of $\mathbb{E}_R(A - e_0 \mid X) = 0$. At $(t, m, e) = (\tau, m_0, e_0)$, replacing e_0 by $e_0 + \varepsilon k$ makes the conditional numerator equal $-\varepsilon^2 k(X)^2 \tau(X)$, proving only local first-order propensity orthogonality.

For each linear sieve, expand its empirical objective and set the penalized gradient to zero. The DRF normal equation is

$$\{(1 - \omega)G_R + \omega G_{O,r} + \rho P\}\beta = h_\lambda + \omega d_g, \quad (80)$$

which proves Equation (61) whenever the matrix is invertible. The RF expansion gives

$$\{(1 - \omega)G_{R,\kappa} + \omega G_{O,r} + \rho P\}\beta = h_{m_\lambda} + \omega d_{g,\kappa},$$

which proves Equation (62).

For Theorem 6, Hoeffding's inequality and a union bound over the M candidates give the uniform deviation ε_{val} . Comparing the selected score with an oracle candidate and cancelling the common Q_{ref} gives the factor two. With approximate weights, direct expansion adds $d_{r,a} + d_{\text{nuis},a}$ to each candidate expectation, so two uniform drift bounds are added.

For Theorem 7, the first-order condition is $(\hat{H}_j + \rho P)\hat{\beta}_j = \hat{h}_j$. Subtracting its value at β_p proves Equation (63). Projection orthogonality, $\mathbb{E}_R[b(t_p - \tau)] = 0$, removes the risk cross term and proves Equation (64).

For Theorem 8, let $\psi_{R,j}$ and $\psi_{O,j}$ be the raw RCT and OBS score contributions. Their source means $a_{R,j} \triangleq P_R \psi_{R,j}$ and $a_{O,j} \triangleq P_O \psi_{O,j}$ need not vanish separately, but exact transport gives $a_{R,j} + a_{O,j} = 0$. Center each source contribution by its own mean. Sub-Gaussian vector concentration then bounds the whitened score by

$$C_{\text{sg}} \left[\sigma_{R,j} \sqrt{\frac{p + \log(4/\delta_S)}{n_f}} + \sigma_{O,j} \sqrt{\frac{p + \log(4/\delta_S)}{N_f}} \right].$$

On $\mathcal{E}_H$, the whitened inverse Hessian has norm at most $(1 - \eta)^{-1}$. The triangle inequality and $(x + y)^2 \leq 2x^2 + 2y^2$ prove Equation (65).

Finally, insert $a_{p_n}^2 = O(K_n^{-2s})$ and $p_n \asymp K_n$ into that bound. The stated triangular-array conditions

make the Gram and score events hold with probability tending to one. Balancing K_n^{-2s} with K_n/m_{eff} proves Corollary 5.

B.11. A two-stratum oracle calculation

Let $P_R(X = 0) = P_R(X = 1) = 1/2$, $\tau(0) = 0$, $\tau(1) = 2$, and $g = \tau$. With $n = 100$, $N = 900$, and a common covariate marginal,

$$\mathbb{E}_R\{\tau(X)\} = 1, \quad \text{Var}_R\{\tau(X)\} = 1, \quad \omega_{\text{ATE}}^* = \frac{1}{1 + 100/900} = 0.9.$$

The baseline between- X variance is 0.01, whereas the oracle component is

$$\frac{\text{Var}_R\{0.1\tau(X)\}}{100} + \frac{0.9^2 \text{Var}_R\{\tau(X)\}}{900} = 0.0001 + 0.0009 = 0.001.$$

The reduction 0.009 concerns only between- X variation; total variance also contains the within- X pseudo-outcome noise from Equation (12).

B.12. Honest variance estimation and evaluation-sample inference

For Proposition 4, fix the coefficients and scores before using the evaluation observations. Ordinary unbiased sample-variance identities give

$$\mathbb{E}\{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)\} = \text{Var}_R(\psi_R), \quad \mathbb{E}\{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)\} = \text{Var}_O(\psi_O).$$

Dividing by n and N , then adding, proves unbiasedness of $\widehat{V}$ by independence of D_R^{eval} and D_O^{eval} .

For Theorem 3, impose the following evaluation regularity conditions after fixing the learned objects:

- (R1) the RCT and OBS evaluation observations are conditionally independent and retain their respective product laws;
- (R2) $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and for some $\delta > 0$ the centered RCT and OBS evaluation-score arrays have uniformly bounded conditional $(2 + \delta)$ moments;
- (R3) $n \text{Var}\{\widehat{\theta}_r(\widehat{\lambda}, \widehat{\omega})\} \rightarrow \sigma^2 \in (0, \infty)$ in probability, and the combined sample-variance estimator is ratio-consistent for this variance.

Center the two evaluation-score arrays at their population means. Their weighted sum is $\widehat{\theta}_r - \theta$. Condition (R2) supplies the two-sample triangular-array central limit theorem: the RCT Lyapunov remainder is of order $n^{-\delta/2}$, and the OBS remainder also vanishes when $n/N \rightarrow \kappa < \infty$. Condition (R3) and Slutsky's theorem yield the studentized limit and asymptotic coverage of the normal-approximation interval.

B.13. Proof of Theorem 4

Under either exact $r = r_0$ transport or its common-marginal special case, Equation (6) gives $\mathbb{E}_R(\Delta \mid X) = 0$. Hence $E = \text{Cov}_R(\Delta, g) = 0$, as shown in Equation (77). Multiplying Equation (16) by n therefore gives

$$\tilde{\mathcal{V}}(\lambda, \omega) = \sigma_0^2 + 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2.$$

At $(\lambda, \omega) = (0, 0)$, this equals σ_0^2 . Holding $\lambda = 0$ and completing the square in ω gives the unique PPI+-only oracle D/B and variance $\sigma_0^2 - D^2/B$. Completing the two separate squares gives the joint oracle $(-C/A, D/B)$ and variance $\sigma_0^2 - C^2/A - D^2/B$. The two variance gaps are D^2/B and C^2/A . Because $A, B > 0$, they are nonnegative and vanish exactly when $D = 0$ and $C = 0$, respectively.

B.14. Density-ratio characterizations and estimated-ratio inheritance

Classifier odds. Let p_R and p_O be densities with respect to a common dominating measure. In the pooled nuisance population,

$$s_0(x) = \frac{\pi p_R(x)}{\pi p_R(x) + (1 - \pi)p_O(x)}.$$

Solving for $p_R(x)/p_O(x)$ gives

$$r_0(x) = \frac{1 - \pi}{\pi} \frac{s_0(x)}{1 - s_0(x)},$$

which proves Equation (29) in Definition 10. The same algebra with a fitted domain probability and the effective training prior yields $\hat{r}_{\text{CLS}}$.

Proof of Proposition 5. The defining property of the Radon–Nikodym derivative gives

$$\mathbb{E}_O\{r_0(X)h(X)\} = \mathbb{E}_R\{h(X)\}$$

for every integrable h . If a nonnegative normalized function $\tilde{r}$ gives the same equality for every bounded measurable h , then $\int_A \tilde{r} dP_O = P_R(A) = \int_A r_0 dP_O$ for every measurable set A . Uniqueness of the Radon–Nikodym derivative gives $\tilde{r} = r_0$ P_O^X -almost surely.

If $r_0 \in L_2(P_O^X)$, then for every $a \in L_2(P_O^X)$,

$$\begin{aligned} \mathcal{R}(a) &= \mathbb{E}_O\{a(X)^2\} - 2\mathbb{E}_R\{a(X)\} \\ &= \mathbb{E}_O\{(a(X) - r_0(X))^2\} - \mathbb{E}_O\{r_0(X)^2\}. \end{aligned}$$

The squared term is uniquely minimized at $a = r_0$ in $L_2(P_O^X)$. Restricting the balance equations to the coordinates of ϕ tests equality only against functions in their linear span, which proves the finite-feature qualification.

Proof of Lemma 1. For $a_\gamma = \phi^\top \gamma$, the gradient of the objective in Equation (31) is twice

$$\left[\mathbb{P}_O^{\text{nuis}} \{ \phi(X) \phi(X)^\top \} + \rho I_p \right] \gamma - \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}.$$

The stated invertibility condition makes its unique zero Equation (32).

For Equation (34), transport identity (27) gives

$$\mathbb{E}_O(g^2 a^2) - 2\mathbb{E}_R(g^2 a) = \mathbb{E}_O\{g^2(a - r_0)^2\} - \mathbb{E}_O(g^2 r_0^2),$$

which proves the claimed weighted excess-risk identity.

Proof of Proposition 6. The log of a positive weighted exponential sum is convex in ξ . At an interior minimizer, differentiating the objective in Definition 12 gives

$$\begin{aligned} 0 &= \frac{\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \phi(X) \}}{\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \}} - \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \} \\ &= \mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{BAL}}(X) \phi(X) \} - \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}. \end{aligned}$$

The denominator normalizes the OBS weights to one, and positivity follows from the positive base ratio and exponential tilt. A finite interior optimizer can exist only when the target feature mean lies in the relative interior of the convex hull of the weighted OBS feature vectors. Boundary or infeasible cases require a separate approximate-balance rule and are not covered by the exact first-order condition.

Proof of Lemma 2. On the tuning OBS sample, $\hat{r}g = r_0g + \delta_{\hat{r}}$. The sample-variance identity $\widehat{\text{Var}}(U + V) = \widehat{\text{Var}}(U) + 2\widehat{\text{Cov}}(U, V) + \widehat{\text{Var}}(V)$ proves Equation (41). When the axis thresholds are inactive,

$$\hat{\omega}_{\hat{r}} - \hat{\omega}_{r_0} = \frac{\hat{D}}{\hat{B}_{\hat{r}}} - \frac{\hat{D}}{\hat{B}_{r_0}},$$

which is Equation (42) after taking a common denominator.

Proof of Proposition 7. Condition on the nuisance and tuning samples. At the same selected coefficients, the only direct difference between $\hat{\theta}_{\hat{r}}$ and $\hat{\theta}_{r_0}$ in Equation (8) is $\hat{\omega} \mathbb{P}_{O,N} \delta_{\hat{r}}$. This proves Equation (43). Adding and subtracting $P_O \delta_{\hat{r}}$ proves Equation (44).

Proof of Corollary 3. Taking the conditional expectation of Equation (44) removes its centered empirical term. The remaining added mean is $\hat{\omega} P_O \delta_{\hat{r}}$, and transport identity (27) gives

$$P_O \delta_{\hat{r}} = \mathbb{E}_O(\hat{r}g) - \mathbb{E}_R(g),$$

which proves Equation (45).

Proof of Corollary 4. Equation (46) makes the scaled population drift in Equation (44) negligible. For the centered OBS evaluation perturbation,

$$\mathbb{E} \left[\left\{ \sqrt{n} \hat{\omega} (\mathbb{P}_{O,N} - P_O) \delta_{\hat{r}} \right\}^2 \right] = \hat{\omega}^2 \frac{n}{N} \|\delta_{\hat{r}} - P_O \delta_{\hat{r}}\|_{L_2(P_O)}^2,$$

so Equation (47) makes this perturbation $o_p(1)$. Tuning-moment consistency and asymptotically inactive thresholds make the selected coefficients converge to their exact-ratio limits. Slutsky's theorem then transfers the normal limit. More explicitly, for the deterministic limit $(\lambda^\circ, \omega^\circ)$,

$$\begin{aligned} \hat{\theta}_{r_0}(\hat{\lambda}, \hat{\omega}) - \hat{\theta}_{r_0}(\lambda^\circ, \omega^\circ) \\ = (\hat{\lambda} - \lambda^\circ) \mathbb{P}_{R,n} \Delta + (\hat{\omega} - \omega^\circ) \{ \mathbb{P}_{O,N}(r_0 g) - \mathbb{P}_{R,n} g \}. \end{aligned}$$

Coefficient consistency makes both coefficient differences $o_p(1)$. The RCT score satisfies $\mathbb{P}_{R,n} \Delta = O_p(n^{-1/2})$, and exact transport gives $\mathbb{P}_{O,N}(r_0 g) - \mathbb{P}_{R,n} g = O_p(N^{-1/2} + n^{-1/2})$. Because n/N is bounded, the entire display is $o_p(n^{-1/2})$. If each deterministic limit is the corresponding exact-ratio population-oracle pair in Theorem 4, this replacement argument transfers its first-order variance ordering. If either ratio-error condition fails, the ratio learner's influence term, the OBS empirical-process term, and their covariances must enter the limiting variance. None of these arguments compares realized finite-sample variances.

B.15. Outer-fold honesty

For Proposition 8, condition on the prespecified or independently randomized split assignments and on one outer complement. Every fitted object is fixed, and the held-out paired folds retain the required RCT and OBS laws. The exact centering argument applies within that fold. Iterated expectation proves aggregate unbiasedness. It does not make different fold statistics independent: in general,

$$\text{Var} \left(\sum_k a_k \hat{\theta}_k \right) = \sum_k a_k^2 \text{Var}(\hat{\theta}_k) + 2 \sum_{k < \ell} a_k a_\ell \text{Cov}(\hat{\theta}_k, \hat{\theta}_\ell).$$

The second sum need not vanish because outer training complements overlap.

B.16. Counterexamples to adaptive reuse

The fixed-coefficient theorem cannot be extended by selecting a coefficient on the same evaluation outcomes without a separate analysis.

Reuse through λ . Let X be degenerate, $\theta = 0$, $Z_0 = 0$, and let the Δ_i be i.i.d. Rademacher variables. Every fixed λ satisfies $\mathbb{E}\{\lambda \mathbb{P}_{R,n} \Delta\} = 0$. If the same outcomes select $\hat{\lambda} = \text{sign}(\mathbb{P}_{R,n} \Delta)$, then

$$\mathbb{P}_{R,n}(Z_0 + \hat{\lambda} \Delta) = |\mathbb{P}_{R,n} \Delta|,$$

whose expectation is strictly positive.

Reuse through ω . Under the common covariate marginal, let $g(X)$ be a nondegenerate mean-zero Rademacher variable and let the RCT-valid score be identically zero. Every fixed ω centers $\omega\{\mathbb{P}_{O,N}g - \mathbb{P}_{R,n}g\}$. Choosing $\hat{\omega}$ as the sign of that same bracket produces its absolute value, which again has strictly positive expectation. The bias in both examples comes solely from adaptive reuse.